Narrative Shocks in the Oil Market: Evidence from 70 Years of News Coverage

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Abstract

This paper studies whether oil-market narratives have macroeconomic effects, and whether those effects differ depending on whether narratives convey directional information or uncertainty. I construct new monthly oil narrative sentiment and uncertainty indices from *Wall Street Journal* and *New York Times* articles from 1953 to 2022 and classify oil-related coverage into five narrative types. Using the type-specific narrative decomposition as external instruments in a Proxy-SVAR, I identify two structurally distinct narrative shocks. A bullish narrative sentiment shock raises the real oil price persistently, increases world industrial production, raises world oil production, and gradually passes through to consumer prices, consistent with a demand-side information channel. By contrast, an elevated narrative uncertainty shock reduces world oil production and world industrial production, while producing only a muted response of the real oil price and consumer prices, consistent with a supply-side real options channel. When both shocks hit simultaneously, the combined effect produces a stagflationary pattern in which oil prices and inflation rise while production and activity respond weakly. These findings show that oil-market narratives contain macroeconomically relevant variation beyond physical oil shocks and that separating directional tone from ambiguity is essential for understanding how oil news transmits to the broader economy.

Keywords: oil-market narratives; sentiment; uncertainty; text-as-data; external instruments; oil-macroeconomy; Proxy-SVAR; stagflation

JEL Classification: C32, E32, Q41, Q43, G14

1 Introduction

Recent turbulence in the oil market has renewed interest in the longstanding question of how oil prices affect the macroeconomy. But oil price movements reflect more than contemporaneous physical disruptions or realized shifts in global demand. They also reflect changing beliefs about future market conditions. News about wars in oil-producing regions, OPEC decisions, speculative activity, refinery outages, and energy-transition policy can alter how market participants interpret future oil scarcity, demand, and investment conditions before any physical barrel changes hands (Kilian and Murphy, 2014; Hamilton, 2009). Yet the macroeconomic consequences of such oil-market news need not be uniform. Some narratives provide directional information that shifts beliefs about future market conditions. Others primarily increase ambiguity about the future path of the oil market. This paper argues that this distinction matters for macroeconomic transmission.

The structural oil-market literature has made substantial progress in distinguishing between supply shocks, global demand shocks, and precautionary or inventory-demand shocks (Kilian, 2009; Kilian and Murphy, 2014; Baumeister and Hamilton, 2019). But these frameworks do not directly separate oil-market news that moves expectations in a directional way from oil-market news that primarily raises ambiguity. As a result, episodes of intense oil-market coverage that differ in economic content may be absorbed into broad residual demand-side or uncertainty-related disturbances. This omission is important because directional narrative shifts and ambiguity shocks imply different adjustment mechanisms, different responses of production and activity, and potentially different interpretations of oil-price movements.

This paper studies whether oil-market narratives have macroeconomic effects, and whether those effects differ depending on whether narratives become more *directional* or more *uncertain*. I distinguish between two dimensions of oil-market news. The first is *narrative sentiment*, which captures whether oil-market coverage is directionally bullish or bearish. The second is *narrative uncertainty*, which captures the degree of ambiguity, unpredictability, or lack of clarity embedded in that coverage. The key hypothesis is that these two dimensions are economically distinct and propagate through different channels. A rise in bullish narrative sentiment may shift expectations about future scarcity or future demand, increasing the willingness of market participants to hold inventories and generating a demand-side information effect (Beaudry and Portier, 2006; Barsky and Kilian, 2002). By contrast, a rise in narrative uncertainty may increase the option value of waiting and delay partially irreversible production and investment decisions (Bernanke, 1983; Dixit and Pindyck, 1994; Bloom, 2009), generating a supply-side delay effect.

To study these mechanisms, I construct new monthly oil-market narrative sentiment and uncertainty indices from *Wall Street Journal* and *New York Times* articles spanning 1953 to 2022. Using a domain-specific oil dictionary, I identify more than 141,000 oil-relevant articles from a raw corpus of nearly two million business and market news articles. Each article is scored separately for directional tone and uncertainty content and then aggregated to the monthly frequency. I also classify oil-related coverage into five narrative types: geopolitical conflict, supply disruption, OPEC coordination, speculation/financialization, and energy transition. This yields both aggregate narrative indices and type-specific measures over seven decades of oil-market history.

A central empirical result is that narrative sentiment and narrative uncertainty are not redundant measures of the same underlying oil-market news factor. They are correlated, but they behave differently across major historical episodes and imply different macroeconomic response patterns. Episodes such as the 1986 OPEC collapse are associated with strongly negative sentiment and little uncertainty, reflecting a market that clearly understood the directional implications of the event.[1] By contrast, episodes such as the Arab Spring exhibit elevated uncertainty alongside bullish sentiment, reflecting bullish supply-risk pricing combined with substantial ambiguity about future geopolitical developments.[2] These patterns suggest that oil-market narratives contain separable information about directional beliefs and ambiguity rather than a single latent news factor.

I then study the macroeconomic effects of these two narrative dimensions using a Proxy-SVAR (Mertens and Ravn, 2013; Stock and Watson, 2018). The identification strategy targets shocks to aggregate oil-market narratives rather than physical oil shocks directly. Five type-specific indices serve as external instruments for the aggregate oil-market narratives. These type-specific indices capture variation from economically distinct sources of oil market news. Geopolitical events, OPEC decisions, speculative positioning, supply disruptions, and energy transition policy each generate narrative variation through different channels. This approach differs from high-frequency event-window designs such as Känzig (2021) and from event-based media-shock instruments such as Gavriilidis et al. (2026).[3] Instead, it exploits heterogeneity within the narrative environment itself. Because identification rests on the plausibility of the underlying components, I report instrument-specific first-stage diagnostics following Montiel Olea et al. (2021), forecastability and serial-correlation tests, over-identification tests (Sargan, 1958; Hansen, 1982), and exclusion exercises.

[1] In late 1985, Saudi Arabia abandoned its role as swing producer and began competing for market share, precipitating a price collapse from roughly $27 to below $10 per barrel in 1986. See Gately (1986) and Hamilton (2013).

[2] The Arab Spring (2010–2012) disrupted production in Libya and raised concerns about contagion to major Gulf producers. See Hamilton (2011) for a discussion of the oil-market implications.

[3] See also Caldara and Iacoviello (2022) for the Geopolitical Risk index and Baker et al. (2016) for the Economic Policy Uncertainty index, both of which use newspaper text to construct aggregate indices but do not decompose them into type-specific instruments for structural identification.

Confidence bands are constructed using the moving block bootstrap following Jentsch and Lunsford (2022).

The empirical results reveal two distinct macroeconomic signatures. A bullish narrative sentiment shock raises the real oil price persistently, increases world industrial production, raises world oil production, and gradually passes through to consumer prices. The positive co-movement of oil prices and activity is difficult to reconcile with an adverse supply interpretation (Kilian, 2009) and is instead consistent with a demand-side information channel in which bullish oil-market narratives shift beliefs about future market tightness or future demand. By contrast, a narrative uncertainty shock reduces world oil production and world industrial production, while producing only a muted and statistically weak response of the real oil price and consumer prices. This pattern is consistent with a supply-side delay channel: heightened ambiguity discourages adjustment in drilling and production, suppressing real activity while leaving the net price effect weak because lower demand and tighter supply work in opposite directions (Bloom, 2009; Elder and Serletis, 2010). Additional evidence from drilling activity supports this interpretation, as narrative uncertainty lowers the Baker Hughes rig count whereas sentiment raises it.

A further implication of the two-channel framework is that simultaneous sentiment and uncertainty shocks can generate a macroeconomic pattern that neither shock produces alone. When bullish oil-market sentiment and elevated uncertainty occur together, oil prices rise because stronger demand expectations and supply concerns move in the same direction, while oil production remains weak because the incentive to expand output is offset by the incentive to delay investment. The result is a stagflationary combination of higher oil prices, subdued production, weaker real activity, and higher inflation. This perspective helps interpret episodes such as the Gulf War (1990–91), the Arab Spring (2011), and the Russia-Ukraine conflict (2022), in which oil-market narratives likely became both more bullish and more uncertain. The standard three-shock framework of Kilian (2009) cannot separate these dimensions because it does not distinguish between the directional and ambiguity components of oil-market news.

These findings imply that oil-market narratives contain macroeconomically relevant variation beyond standard physical oil shocks and that separating directional tone from ambiguity is important for understanding how oil news transmits to the broader economy. More broadly, the results suggest that standard oil-shock frameworks (Kilian, 2009; Baumeister and Hamilton, 2019) may conflate two distinct forms of oil-market news: one that behaves like a demand-side expectations shock and another that behaves more like a supply-side uncertainty shock. Distinguishing between them helps explain why some episodes of intense oil-market coverage generate price-output co-movement, while others primarily suppress production and activity without producing a comparably strong price response.

The paper contributes to several literatures. First, it contributes to the structural oil-market literature following Kilian (2009) by showing that oil-market narratives contain macroeconomically relevant variation not well summarized by physical supply, global demand, or inventory-demand shocks alone. Second, it contributes to narrative economics (Shiller, 2017) by showing that a single domain-specific news corpus can recover distinct dimensions of narrative content with different macroeconomic consequences. Third, it contributes to the growing literature using text-based measures to identify economically meaningful shocks (Baker et al., 2016; Caldara and Iacoviello, 2022; Shapiro et al., 2022; Larsen and Thorsrud, 2019) by showing how decomposition within a corpus can generate multiple proxies for a common aggregate narrative disturbance. In this sense, the paper is not only about measuring oil-market narratives, but also about identifying how different forms of narrative variation map into different macroeconomic transmission channels.

The paper also contributes methodologically. Many text-based measures are treated as reduced-form indicators whose economic content is difficult to interpret. The approach here shows how decomposition within a corpus can be used to generate multiple proxies for a common aggregate narrative shock, conceptually related to multi-instrument designs in the IV literature (Angrist and Krueger, 1991; Stock and Yogo, 2005). In this respect, the strategy is conceptually related to multi-component instrument designs: identification hinges not on a single black-box index, but on the plausibility of the underlying components and on whether the results are robust to excluding problematic ones. In this application, the diagnostics indicate strong first-stage relevance in both systems and broadly support the validity of the type-specific instruments, while also revealing that OPEC sentiment contains additional price-relevant content beyond common aggregate sentiment. Reassuringly, excluding that instrument leaves the qualitative results unchanged.

The remainder of the paper proceeds as follows. Section 2 develops the conceptual framework. Section 3 describes the newspaper corpus, article scoring, and narrative classification. Section 4 presents the Proxy-SVAR and the instrument construction. Section 5.1, Section 5.2, Section 5.3, and Section 5.4 report the main empirical results, followed by historical decompositions, variance decompositions, and robustness checks. The final section concludes.

2 Conceptual Framework

This section develops the economic logic underlying the paper's central hypothesis: oil-market narratives can affect the macroeconomy through two distinct dimensions of news content. The first is *narrative sentiment*, which captures the directional tone of oil-market coverage. The second is *narrative uncertainty*, which captures the ambiguity or unpredictability embedded in that coverage. The key premise of the paper is that these two dimensions need not transmit to the economy through the same channel.

The structural oil-market literature, beginning with Kilian (2009), emphasizes shocks originating in the *physical* oil market: realized supply disruptions, shifts in global demand, and changes in precautionary demand driven by concerns about future scarcity (see also Kilian and Murphy, 2014; Baumeister and Hamilton, 2019). These shocks operate through observable market outcomes such as production, consumption, inventories, and prices. But the same events that move the physical oil market also generate intense news coverage, and that coverage may itself shape beliefs and decisions before any physical quantity adjusts. A conflict in the Middle East, for example, is both a potential source of supply disruption and a source of narratives that may alter expectations about future scarcity, volatility, and investment conditions (Hamilton, 2003).

Existing frameworks do not directly distinguish between oil-market news that shifts directional expectations and oil-market news that primarily raises ambiguity.[4] As a result, episodes of intense oil-market coverage that differ in economic content may be absorbed into broad residual demand-side or uncertainty-related disturbances. This matters because directional belief shifts and ambiguity shocks imply different propagation mechanisms, especially for investment and production decisions. This paper studies whether such narrative variation has macroeconomic effects that are distinct from, and additional to, the physical channel.

The paper does not identify physical oil shocks directly. Instead, it identifies shocks to aggregate oil-market narratives and interprets their macroeconomic effects through the joint behavior of oil prices, oil production, real activity, and consumer prices. The conceptual framework clarifies two points. First, it motivates why sentiment and uncertainty should be treated as distinct dimensions of oil-market narratives. Second, it delivers testable predictions for how each dimension should affect oil prices, production, and real activity, which guide the empirical analysis.

2.1 Narrative Sentiment and the Demand-Side Information Channel

Narrative sentiment captures the directional framing of oil-market conditions. When coverage becomes more bullish, emphasizing stronger demand, tighter supply, recovering activity, or sustained price pressure, market participants may revise upward their expectations of future oil scarcity or future oil demand. This mechanism is closely

[4] The closest precedents are Jo (2014), who distinguish oil-price uncertainty from oil supply shocks, and Baumeister et al. (2018), who separate expected and unexpected components of energy-price changes. Neither, however, extracts distinct narrative dimensions from text data.

related to the oil-specific demand channel in Kilian (2009): forward-looking agents become more willing to hold inventories and bid up current prices in anticipation of tighter conditions ahead.

Importantly, this channel does not require that media narratives contain fundamentally new information in the strict rational expectations sense. Following the logic of narrative economics (Shiller, 2017), news coverage can shape expectations by altering salience, framing, and repetition. This is consistent with models of limited attention (Sims, 2003) and information rigidities (Mankiw and Reis, 2002), in which agents update beliefs gradually and the framing and salience of information sources matter for the speed and direction of expectation revision. A newspaper article about record speculative long positions in crude oil, for example, may not reveal facts that are unavailable elsewhere, but it may still change how those facts are interpreted and weighted by market participants (Tetlock, 2007). At the aggregate level, repeated shifts in the directional tone of oil-market coverage can therefore move beliefs about future market conditions and generate macroeconomically relevant information shocks.

This reasoning implies a clear empirical signature. If a bullish narrative sentiment shock operates primarily through a demand-side information channel, it should raise the real price of oil and, at least over the relevant horizon, move oil prices and real activity in the same direction. Oil production may also rise over time as producers respond endogenously to stronger price incentives, and consumer prices may increase gradually through oil-price pass-through (Hamilton, 2013; Blanchard and Galí, 2007).

2.2 Narrative Uncertainty and the Supply-Side Delay Channel

Narrative uncertainty captures the degree of ambiguity surrounding future oil-market conditions. When coverage emphasizes conflicting signals, unclear outcomes, geopolitical instability, or unpredictable policy and market developments, the narrative environment may widen the perceived range of future oil-market outcomes. Such a rise in uncertainty can matter even when the directional implications of the news remain unclear.

A natural mechanism linking this uncertainty to macroeconomic outcomes is the wait-and-see logic developed by Bernanke (1983) and Bloom (2009). When decisions are costly to reverse, greater uncertainty raises the option value of waiting. In the oil sector, this logic is especially relevant because drilling, exploration, and infrastructure investment are capital-intensive and partly irreversible (Dixit and Pindyck, 1994). Kellogg (2014) shows that uncertainty about future oil prices can reduce drilling activity, consistent with this mechanism.[5] This mechanism implies that narrative uncertainty

[5] Agerton et al. (2017) extend this evidence by documenting that oil-price uncertainty affects employment decisions in U.S. oil-producing counties. Elder and Serletis (2010) find that oil-price uncertainty

operates as a supply-side disturbance, not because it directly reduces productive capacity, but because it delays the adjustment of partially irreversible production decisions.

In the present context, the idea is not that narrative uncertainty is identical to firm-level investment uncertainty, but that heightened ambiguity in oil-market narratives may generate a macroeconomic environment in which producers delay action. If so, a narrative uncertainty shock should depress oil production persistently, weaken real activity, and exert only an ambiguous effect on the real oil price. The reason is that two forces operate in opposite directions: weaker activity reduces oil demand and pushes prices down, while delayed production and investment constrain supply and push prices up. Consumer prices may therefore respond only modestly, if at all.

2.3 Two Dimensions of the Same Narrative Environment

A central feature of the paper is that both dimensions are extracted from the same newspaper corpus. The same article can contain directional content and uncertainty content at the same time. For example, a report on an OPEC production cut may be bullish in tone for oil prices while also emphasizing uncertainty about compliance, duration, or the broader geopolitical response. Conversely, coverage can be sharply directional with little ambiguity, or highly uncertain without a strong directional signal.[6]

This distinction is useful because it implies that sentiment and uncertainty are not simply alternative labels for the same narrative shock. They describe different features of the narrative environment. The empirical analysis therefore treats them as separate aggregate objects and asks whether shocks to each dimension generate different macroeconomic response patterns.

Table 1 illustrates the distinction. The four quadrants combine directional tone (bullish or bearish) with the level of ambiguity (low or high). The examples are stylized, but they capture the basic idea that narrative tone and narrative uncertainty can vary independently.

2.4 Predicted Macroeconomic Signatures

The framework yields a set of qualitative predictions that guide the empirical analysis.

First, a bullish narrative sentiment shock should be associated with a rise in the real price of oil. If the dominant mechanism is demand-side information, real activity

depresses aggregate investment and output.

[6]This feature distinguishes the approach from designs that extract a single latent factor from text, such as the news sentiment index of Shapiro et al. (2022) or the topic-based approach of Larsen and Thorsrud (2019). Here, two distinct dimensions are scored from each article.

Table 1: Illustrative Combinations of Sentiment and Uncertainty in Oil-Market Narratives

	Low uncertainty	**High uncertainty**
Bearish sentiment	"OPEC's decision to flood the market sent prices into freefall; analysts say the cartel has abandoned supply management."	"Markets fear severe disruption from the spreading conflict, but the magnitude and duration of any supply loss remain deeply unclear."
Bullish sentiment	"Output has stabilized and inventories are rebuilding steadily as demand from China continues to exceed forecasts."	"Producers expect a strong recovery in demand, though the pace and durability of the rebound remain highly uncertain."

Note: The examples are stylized composites of oil-market reporting and illustrate the distinction between the directional tone of coverage and the ambiguity surrounding that coverage. The scoring procedure described in Section 3.3 extracts these two dimensions separately from each article.

should also increase, producing positive co-movement between oil prices and output. Oil production may rise with a delay as producers respond to stronger price incentives, and consumer prices may increase gradually through pass-through.

Second, a high narrative uncertainty shock should be associated with a persistent decline in oil production if heightened ambiguity induces delay in irreversible decisions. Real activity should weaken, while the response of the real oil price should be muted or ambiguous because the downward demand effect and the upward supply effect work in opposite directions. The response of consumer prices should therefore be smaller and less precisely signed than under a sentiment shock.

Third, the framework also generates a prediction for episodes in which sentiment and uncertainty rise together. When bullish narrative sentiment coincides with elevated narrative uncertainty, the demand-side information channel pushes oil prices upward while the supply-side delay channel restrains production. Because the two forces operate on different margins, the combined effect is a stagflationary pattern in which oil prices and inflation rise, while production and real activity respond weakly relative to a pure sentiment shock. This joint response is not implied by either shock in isolation. Section 5.4 examines this prediction directly.

These predictions provide a disciplined benchmark for interpreting the empirical results. If sentiment shocks operate through a demand-side information channel, they should generate positive co-movement between oil prices and real activity. If uncertainty shocks operate through a supply-side delay channel, they should suppress production and activity while producing a weaker and less precisely signed price response. The empirical analysis evaluates whether the data exhibit these distinct patterns.

Figure 1 summarizes the two channels. The top path links narrative sentiment to a demand-side information mechanism. The bottom path links narrative uncertainty to a supply-side delay mechanism consistent with wait-and-see behavior. The empirical analysis asks whether the estimated macroeconomic responses follow these distinct qualitative patterns.

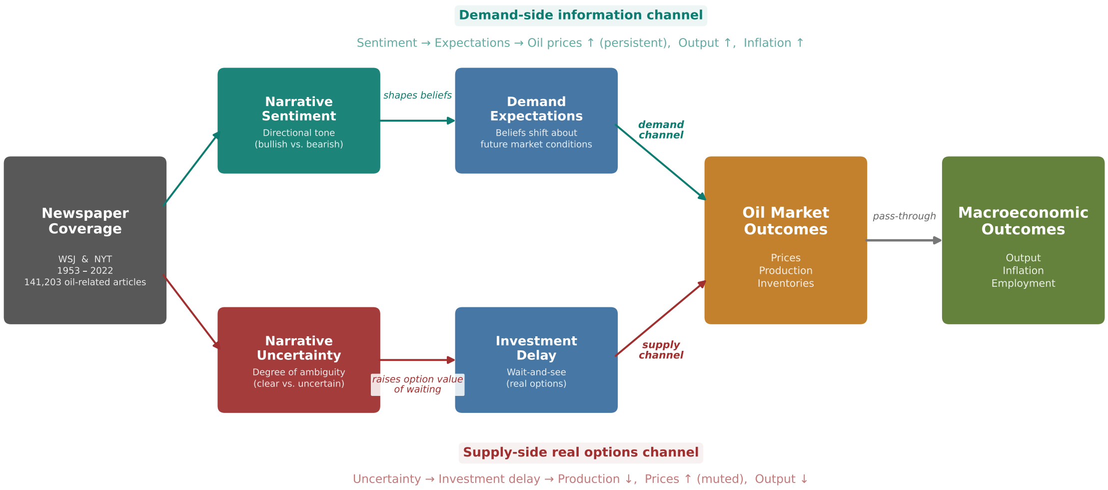


Figure 1: Transmission mechanism: two channels from oil-market narratives to the macroeconomy. The top path traces the demand-side information channel: narrative sentiment shapes expectations about future market conditions, raising oil prices, output, and inflation. The bottom path traces the supply-side uncertainty channel: narrative uncertainty raises ambiguity, delays irreversible decisions, and suppresses oil production with a weaker and less directional price response.

3 Data

This section describes the construction of the narrative indices. The process involves three steps: (i) assembling the newspaper corpus, (ii) scoring articles for sentiment and uncertainty content, and (iii) classifying articles by narrative types.

3.1 Text Corpus

The text data are drawn from ProQuest TDM Studio. We collect all articles from the Business, Finance, Economy, and Markets sections of the *Wall Street Journal* and *New York Times* between January 1953 and December 2022, using section-level metadata rather than keyword searches to avoid selection bias (Gentzkow et al., 2019). The raw corpus comprises 1,973,464 articles; after removing 1,146 with missing text, the final corpus contains 1,972,318 articles spanning 840 months.

3.2 Oil/Energy Article Identification

We identify oil-relevant articles using a domain-specific dictionary organized into five categories: direct oil references ("crude oil," "petroleum," "WTI"), OPEC/geopolitics ("OPEC," "oil embargo"), upstream ("oil drilling," "shale oil," "fracking"), downstream ("refinery," "gasoline price"), and broader energy ("energy crisis," "energy price"). An article is classified as oil-relevant if it contains at least one core oil term or at least two terms from any category combination, yielding 141,203 articles (7.2%). Figure 2 provides a sample oil-related news article in the corpus.

IN THE MARKETS

OPEC Members Inch Toward Oil-Supply Cuts

VIENNA—OPEC members are inching toward a compromise that could lead them to cut oil supply, as the producer group this week prepares for one of its most closely watched meetings in years.

By Benoît Faucon, Summer Said and Sarah Kent

Saudi Arabia, the de facto leader of the Organization of the Petroleum Exporting Countries, is likely to side with calls for the group to adhere more closely to its self-imposed production ceiling at Thursday's meeting of OPEC oil ministers, according to a Gulf official familiar with the Saudi position.

COMMODITIES

Support for such a move, which would be based on tighter compliance with OPEC's existing output limit as opposed to an outright cut to its production target, was coalescing last week at a meeting of OPEC advisers, according to several of those who were present. With oil prices having recently stabilized at around $80 a barrel following a 30% slide since the summer, some within OPEC have seen less urgent need for the group to take stronger action.

But as late as last week, the Saudis' likely approach at Thursday's meeting was unclear to most members. "Saudi Arabia will likely go for a rollover" of the current ceiling, the Gulf official said on Tuesday. A call for strict compliance would be "an option they would go for." But Riyadh's intentions haven't been clearly telegraphed to other members, and it could decide to go a different route Thursday.

OPEC members have engaged in an unusually frantic round of shuttle diplomacy ahead of this week's gathering, with some poorer countries calling for deep supply cuts to boost oil prices. Brent crude ended 1.7% lower at $78.33 a barrel on Tuesday after a meeting among Saudi Arabia, Venezuela and non-OPEC members Russia and Mexico resulted in no agreement to cut oil production. The benchmark crude-oil contract on the New York Mercantile Exchange fell 2.2% to $74.09 a barrel.

OPEC three years ago agreed to limit its collective output to 30 million barrels of oil a day. But its actual production has regularly spilled over as each member has pushed to maintain its share of global oil markets. Closer compliance would imply a supply cut of around 300,000 barrels of oil a day from October levels, according to OPEC technocrats who met last week.

By some distance OPEC's largest oil producer, Saudi Arabia has been unusually reticent in the run-up to this week's meeting. Its oil minister, Ali al-Naimi, refused to answer questions about the country's stance when he arrived in Vienna on Monday.

Earlier this month, Mr. al-Naimi told Venezuela's foreign minister, Rafael Ramírez, that the kingdom wouldn't cut its oil output alone, according to people familiar with the matter.

But a supply cut of around 300,000 barrels a day is unlikely to satisfy some OPEC members, such as Venezuela and Iran, who need oil prices well above $100 a barrel to balance their government budgets.

An OPEC production cut of around 1.5 million barrels would be needed to tighten global oil markets next year, according to oil analysts at Barclays.

Current oil prices "aren't acceptable, of course not," Iraq's oil minister Adel Abdul-Mehdi told reporters upon arriving in Vienna. "A means should be used to raise the prices," he said.

But officials from Persian Gulf countries are concerned that deep output cuts would prove self-defeating since a rise in oil prices would only encourage further U.S. shale-oil production. OPEC members could then lose market share, one Gulf oil official said.

UAE energy minister Suhail Mohamed Al Mazrouei in Vienna on Tuesday.

Agence France-Presse/Getty Images

Figure 2: Sample news article from the *Wall Street Journal*. This illustrates the type of oil narrative content in the corpus

3.3 Text Processing and Scoring

Each oil-related article i is scored along two dimensions, *sentiment* and *uncertainty*, using the formulae:

$$s_i = \frac{N_i^+ - N_i^-}{N_i^{\text{total}}}, \qquad u_i = \frac{N_i^{\text{unc}}}{N_i^{\text{total}}}. \tag{1}$$

Here, N_i^+ and N_i^- denote the counts of positive and negative lexicon words in article i, N_i^{unc} denotes the count of uncertainty-related words, and N_i^{total} is the total word count. The sentiment dictionary is based on the Loughran and McDonald (2011) financial sentiment lexicon, supplemented with a small number of oil-specific terms that are unambiguous in context (e.g., "glut" as negative, "rally" as positive). The uncertainty dictionary draws on Baker et al. (2016) and Loughran and McDonald (2011) and includes terms such as "uncertain," "risk," "unpredictable," "unclear," and related variants.[7]

[7] The distinction between ambiguity (Knightian uncertainty) and risk is formally developed in Ellsberg (1961) and Gilboa and Schmeidler (1989). While the narrative uncertainty index constructed here does not map directly onto the formal ambiguity-aversion framework, the underlying logic is

An important clarification: the sentiment index measures the directional tone of oil market coverage from the perspective of *market conditions*, whether the narrative environment is bullish (prices rising, demand strong, supply tightening) or bearish (prices falling, demand weak, oversupply). A bullish sentiment score does not mean "good news for consumers" or "good news for producers"; it means that the narrative framing of oil market conditions is directionally positive. This is the relevant dimension for identification because it is the bullish or bearish framing of media coverage and not the welfare implications for any particular agent, that shapes the demand expectations through which the information channel operates. The same convention is used by Shapiro et al. (2022) for their news sentiment index and by Bybee et al. (2024) for business news topics.

We aggregate article-level scores to monthly frequency by averaging across all oil-relevant articles published in each calendar month, weighting each article equally: $\text{Sent}_t = M_t^{-1} \sum_{i=1}^{M_t} s_{i,t}$ and $\text{Unc}_t = M_t^{-1} \sum_{i=1}^{M_t} u_{i,t}$, where M_t is the number of oil-relevant articles in month t. The resulting monthly series are standardized to zero mean and unit variance over 1953:01–2022:12. Figure 3 displays the two aggregate indices over the full sample.

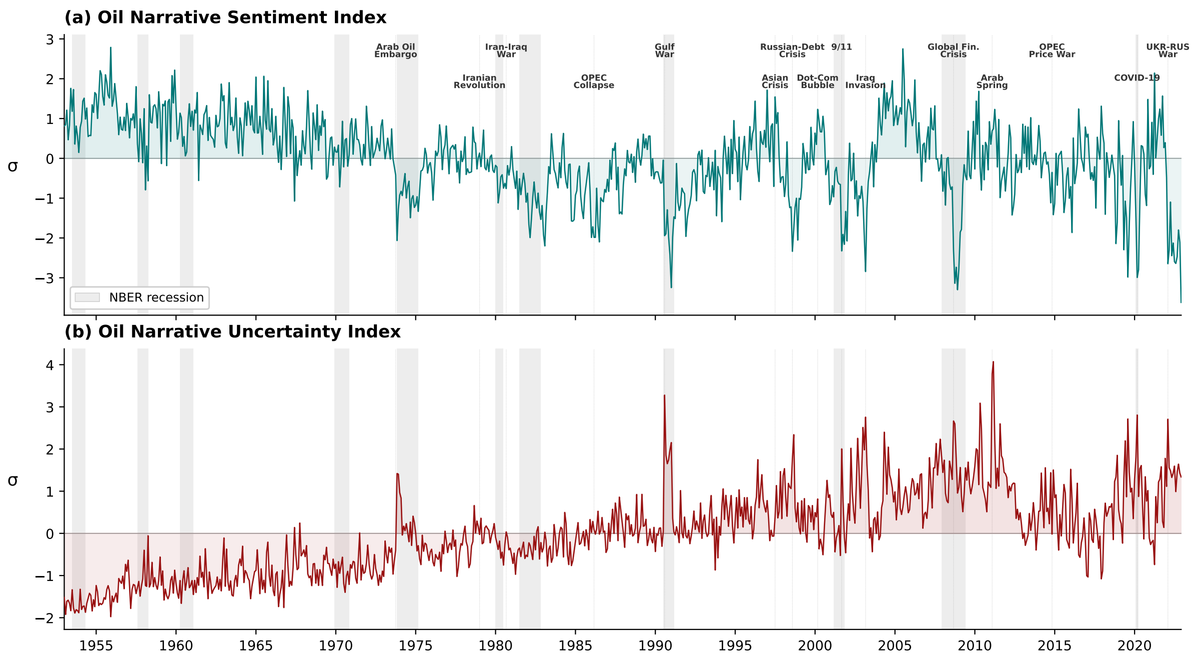


Figure 3: Oil narrative sentiment and uncertainty indices, 1953:01–2022:12. Panel (a): oil narrative sentiment index. Panel (b): oil narrative uncertainty index. Both indices are standardized to zero mean and unit variance over the full sample. Shaded vertical bars denote NBER recession dates. Dotted vertical lines mark major oil market events.

Several features of Figure 3 merit discussion. First, the sentiment index (panel a) ex-

related: elevated narrative uncertainty reflects situations in which agents face a wider and harder-to-characterize range of future oil-market states.

hibits a secular decline from the 1950s through the 2000s, reflecting the transition from a stable, administered oil market under the Texas Railroad Commission regime to the volatile, financialized market of the post-deregulation era (Adelman, 1995; Hamilton, 2013). The deepest troughs correspond to well-known episodes of oil market distress: the Arab Oil Embargo (-2.07σ), the OPEC collapse of 1986 (-1.99σ), 9/11 (-2.33σ), the Global Financial Crisis (-2.73σ), COVID-19 (-2.80σ), and the Russia-Ukraine war (-3.62σ, the sample minimum). Second, the uncertainty index (panel b) shows the opposite secular pattern: persistently below average before 1973 and structurally elevated afterward, with the largest spike occurring during the Arab Spring ($+4.07\sigma$), followed by 9/11 ($+2.00\sigma$) and the Global Financial Crisis ($+1.77\sigma$). Third, while the two indices are negatively correlated ($\rho = -0.46$), they are far from redundant. The OPEC collapse of 1986 illustrates the distinction: sentiment plunges (-1.99σ) as coverage turns overwhelmingly bearish, yet uncertainty barely moves (-0.02σ) because the market understood exactly what was happening (OPEC was flooding supply). Conversely, the Arab Spring generates the sample's largest uncertainty spike ($+4.07\sigma$) alongside *bullish* sentiment ($+0.74\sigma$), reflecting extreme ambiguity about geopolitical outcomes co-existing with bullish supply-risk pricing. These patterns, and especially the divergences between the two series, motivate the central identification strategy: sentiment and uncertainty capture distinct dimensions of narrative content that transmit to the macroeconomy through different channels.

3.4 Narrative Classification

Each oil-relevant article is assigned to a dominant narrative type based on keyword matching against five domain-specific sub-dictionaries: (i) *geopolitical conflict* (wars, sanctions, military intervention), (ii) *supply disruption* (outages, hurricanes, pipeline failures), (iii) *OPEC coordination* (quotas, production cuts, cartel dynamics), (iv) *speculation/financialization* (futures trading, hedge funds, index investment), and (v) *energy transition* (climate policy, renewables, electric vehicles). Articles not matching any sub-type at a threshold of two keyword occurrences are classified as *general oil*. Title matches are weighted three times body-text matches to reflect the salience of headline framing.

Table 2 reports the classification distribution. The largest classified type is speculation/financial (10.1%), followed by geopolitical (4.4%), OPEC (2.8%), supply disruption (1.8%), and energy transition (1.7%). The remaining 79.2% are general oil articles, as expected, since the sub-dictionaries prioritize precision over recall. Monthly type-specific sentiment and uncertainty indices are computed as in Section 3.3, standardized over the estimation sample.[8]

[8] Appendix A validates both the oil-relevance filter and classification using large language model.

Table 2: Narrative Type Distribution of Oil-Relevant Articles

Narrative Type	Articles	Share (%)
Speculation / Financial	14,312	10.1
Geopolitical Conflict	6,241	4.4
OPEC Coordination	3,902	2.8
Supply Disruption	2,505	1.8
Energy Transition	2,395	1.7
General Oil	111,848	79.2
Total	141,203	100.0

Notes: Classification based on keyword matching against five sub-dictionaries. Threshold: ≥ 2 matches. Titles weighted $3\times$.

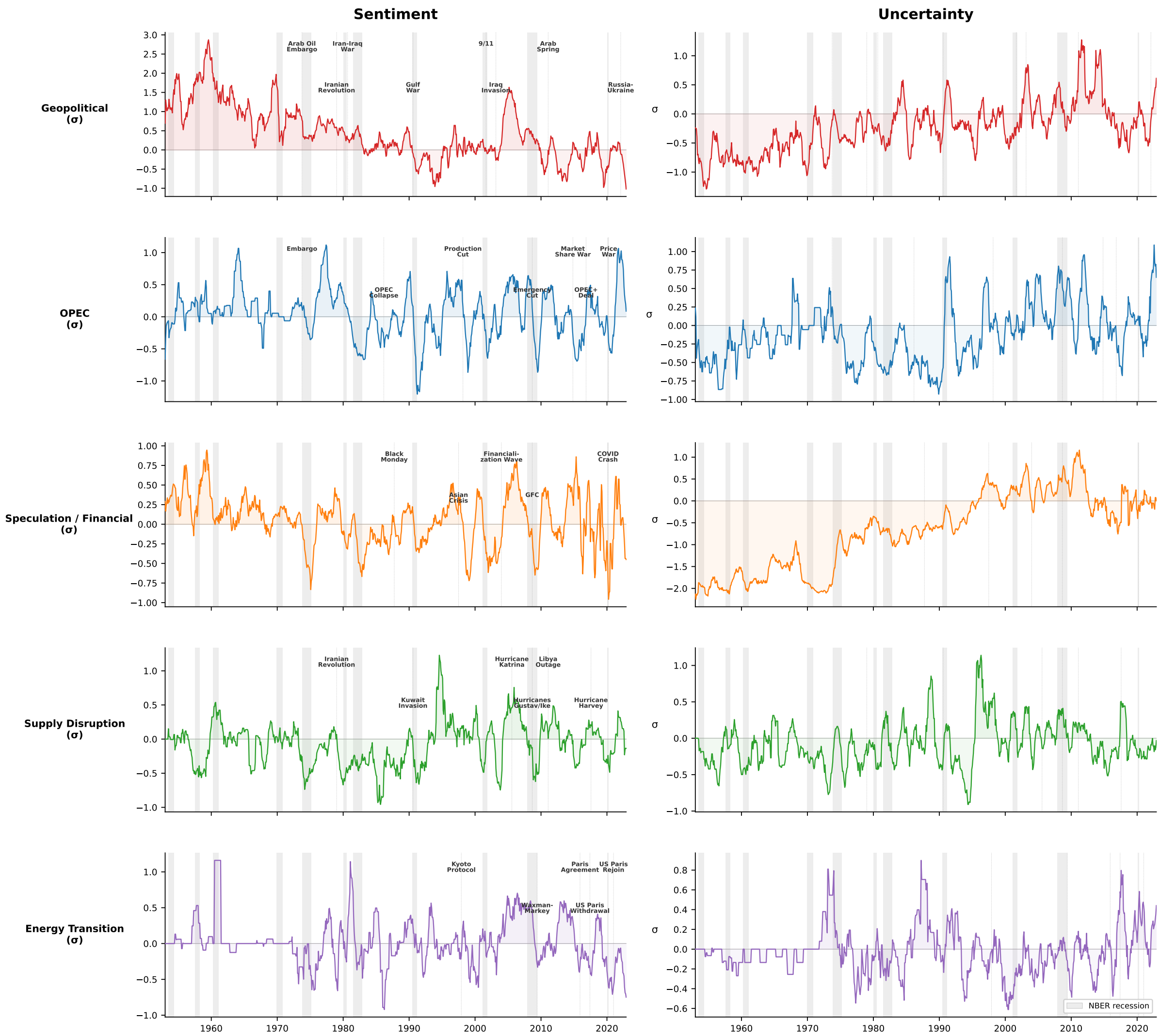


Figure 4: Type-specific narrative indices, 1953–2022. Each row corresponds to one narrative type: geopolitical conflict, OPEC coordination, speculation/financial, supply disruption, and energy transition. Left column: sentiment (directional tone). Right column: uncertainty (degree of ambiguity). All series are 12-month moving averages of the standardized type-specific indices. Dotted vertical lines mark type-relevant events. Shaded vertical bars denote NBER recession dates.

Figure 4 displays the evolution of narrative type shares over the full 1953–2022 sample. The patterns track oil market history closely: speculation narratives surge after 2004 (the financialization wave), OPEC narratives spike during every major coordination episode, geopolitical narratives peak during the Gulf War and Iraq invasion, and energy transition narratives emerge after the 2015 Paris Agreement and accelerate sharply after 2020.

3.5 Index Validation

We assess the quality of the oil narrative indices along two dimensions: narrative plausibility and distinctness from existing measures.

Narrative plausibility. As documented in the discussion of Figures 3 and 4, both the aggregate and type-specific indices align closely with qualitative accounts of major oil market episodes. At the aggregate level (Figure 3), sentiment troughs coincide with well-known episodes of oil market distress, such as the Arab Oil Embargo (-2.07σ), the Global Financial Crisis (-2.73σ), COVID-19 (-2.80σ), and the Russia-Ukraine war (-3.62σ), while uncertainty spikes align with periods of genuine ambiguity about oil market outcomes, such as the Arab Spring ($+4.07\sigma$), 9/11 ($+2.00\sigma$), and the Global Financial Crisis ($+1.77\sigma$). At the type-specific level (Figure 4), each index responds to type-relevant events: geopolitical sentiment deteriorates sharply during armed conflicts, OPEC sentiment swings around coordination episodes, speculation sentiment tracks the financialization wave of the mid-2000s, supply disruption uncertainty rises during hurricanes, and energy transition coverage emerges after the Paris Agreement and accelerates after 2020. Crucially, the type-specific indices respond to *different* events at *different* times, during the Gulf War (1990–91), geopolitical sentiment plunges while speculation sentiment remains near zero; during the financialization wave (2004–08), speculation uncertainty rises sharply while supply uncertainty remains subdued. This cross-type heterogeneity is precisely the variation that the identification strategy exploits.

Distinctness from existing measures. Table 3 reports pairwise correlations between the oil narrative indices and three widely used measures: the University of Michigan Consumer Sentiment Index (UMCSent), the Caldara and Iacoviello (2022) Geopolitical Risk index (GPR), and the Abiad and Qureshi (2023) Oil Policy Uncertainty index (OPU).

Panel A examines the aggregate indices. Oil narrative sentiment is positively correlated with consumer sentiment ($\rho = +0.30$), negatively correlated with geopolitical risk ($\rho = -0.32$), and weakly correlated with oil policy uncertainty ($\rho = -0.17$). These signs are economically intuitive: bullish oil market coverage tends to coincide with higher consumer confidence and lower geopolitical tension. Oil narrative uncertainty is negatively correlated with consumer sentiment ($\rho = -0.21$) and positively correlated with both geopolitical risk ($\rho = +0.21$) and oil policy uncertainty ($\rho = +0.30$). The cross-correlation between the two narrative indices ($\rho = -0.46$) confirms the pattern documented in the discussion of Figure 3: sentiment and uncertainty are related but distinct, with episodes such as the OPEC collapse of 1986 and the Arab Spring of 2011 demonstrating that bearish tone and elevated ambiguity can move independently.

Table 3: Correlation of Narrative Indices with Other Measures

	UMCSent	GPR	OPU	Sent/Unc
Panel A: Aggregate indices				
Oil narrative sentiment	$+0.30$	-0.32	-0.17	-0.46
Oil narrative uncertainty	-0.21	$+0.21$	$+0.30$	—
N	631	456	648	840
Panel B: Type-specific indices				
	Sentiment dimension			
Geopolitical	$+0.11$	-0.06	-0.04	
OPEC	$+0.07$	-0.17	-0.09	
Speculation	$+0.12$	-0.13	-0.03	
Supply	$+0.10$	-0.07	-0.09	
Transition	$+0.03$	-0.07	$+0.01$	
	Uncertainty dimension			
Geopolitical	-0.14	$+0.16$	$+0.11$	
OPEC	-0.09	$+0.14$	$+0.12$	
Speculation	-0.07	$+0.07$	$+0.17$	
Supply	$+0.07$	$+0.05$	$+0.03$	
Transition	-0.05	$+0.10$	$+0.01$	

Notes: Pairwise correlations between the oil narrative indices and other widely used measures. UMCSent: University of Michigan Consumer Sentiment Index. GPR: Caldara and Iacoviello (2022) Geopolitical Risk index. OPU: Abiad and Qureshi (2023) Oil Policy Uncertainty index. Sent/Unc: cross-correlation between the two aggregate narrative indices. Panel A reports correlations for the aggregate (standardized) sentiment and uncertainty indices. Panel B reports correlations for the ten type-specific indices that serve as instruments in the Proxy-SVAR. Correlations computed over the maximal overlapping sample for each pair. N reports the number of monthly observations.

Panel B reveals the key advantage of the narrative type decomposition. While the aggregate indices show moderate correlations with standard measures (Panel A), the type-specific indices are nearly orthogonal to all three comparison measures. The largest type-specific correlation in the entire panel is -0.17 (OPEC sentiment with GPR), and most fall below $|0.10|$. This near-orthogonality arises because the type decomposition strips out the common component that drives the aggregate correlations and isolates source-specific narrative variation (geopolitical events, OPEC decisions, speculative activity, physical supply disruptions, and energy transition policy) that no existing index captures at this level of granularity. Two examples sharpen the point. Supply disruption uncertainty, which captures ambiguity related to hurricanes, infrastructure damage, and pipeline outages, correlates just $+0.05$ with geopolitical risk and $+0.03$ with oil policy uncertainty (physical supply events have essentially nothing to do with geopolitics or policy). Energy transition sentiment, which captures the directional tone of climate and renewables coverage, correlates just $+0.03$ with consumer sentiment and -0.07 with geopolitical risk. This means that, the structural transformation of the energy sector operates on a different dimension entirely from business-cycle sentiment or political instability.

These results establish that the oil narrative indices, and especially the type-specific instruments, capture economically meaningful variation that is distinct from existing measures of sentiment, uncertainty, and risk.

4 Econometric Framework

4.1 Proxy-SVAR Identification

We model the joint dynamics of narrative and macroeconomic variables using a reduced-form VAR of order p:

$$\boldsymbol{Y}_t = \boldsymbol{\mu} + \sum_{\ell=1}^{p} \boldsymbol{B}_\ell \boldsymbol{Y}_{t-\ell} + \boldsymbol{u}_t, \qquad \boldsymbol{u}_t \sim (0, \boldsymbol{\Sigma}_u), \tag{2}$$

where $\boldsymbol{Y}_t$ is an $n \times 1$ vector of endogenous variables with $n = 6$. The structural relationship is $\boldsymbol{u}_t = \boldsymbol{A}_0^{-1}\boldsymbol{\varepsilon}_t$, where $\boldsymbol{\varepsilon}_t$ is the vector of mutually uncorrelated structural shocks and $\boldsymbol{\Sigma}_u = \boldsymbol{A}_0^{-1}(\boldsymbol{A}_0^{-1})'$.

Identification of a single structural shock follows the Proxy-SVAR framework of Mertens and Ravn (2013) and Stock and Watson (2018). Let $\varepsilon_{1,t}$ denote the structural shock of interest and let $\mathbf{Z}_t$ denote the $k \times 1$ vector of external instruments ($k = 5$). The instruments must satisfy:

$$E[\mathbf{Z}_t \varepsilon_{1,t}] = \boldsymbol{\phi} \neq 0 \quad \text{(relevance)}, \tag{3}$$

$$E[\mathbf{Z}_t \varepsilon_{j,t}] = 0 \quad \text{for } j \neq 1 \quad \text{(exogeneity)}. \tag{4}$$

The first-stage regression tests relevance by projecting the reduced-form residual of the instrumented variable, $u_{1,t}$, on the instruments:

$$u_{1,t} = \mathbf{Z}_t'\gamma + v_t. \tag{5}$$

The F-statistic from (5) serves as the primary diagnostic for instrument strength. The fitted values $\hat{u}_{1,t}$ recover the structural impulse vector $\mathbf{b}_1$ (the column of A_0^{-1} corresponding to $\varepsilon_{1,t}$) up to scale and sign (Mertens and Ravn, 2013). Because $k = 5 > 1$, the system is over-identified, which strengthens the first stage relative to any single instrument.

We estimate two separate systems. In the **sentiment system**, the shock of interest is the narrative sentiment shock ($\varepsilon_{1,t} = \varepsilon_t^{\text{sent}}$), and the five sentiment-type instruments serve as $\mathbf{Z}_t$. In the **uncertainty system**, the shock of interest is the narrative uncertainty shock ($\varepsilon_{1,t} = \varepsilon_t^{\text{unc}}$), and the five uncertainty-type instruments serve as $\mathbf{Z}_t$. Importantly, both aggregate indices (*sentiment* and *uncertainty*) enter both VARs as endogenous variables. This allows us to test whether the sentiment shock moves the uncertainty index and vice versa, providing direct evidence on the orthogonality of the two identified shocks.

Confidence bands are constructed using the moving block bootstrap (MBB) of Stock and Watson (2018) with block length $\lceil 5.03T^{1/4} \rceil$, where T is the effective sample size. We report 90% and 68% confidence intervals based on 10,000 bootstrap replications. Forecast error variance decompositions and historical decompositions are computed following Kilian and Lütkepohl (2017). As a specification check, we also report impulse responses from local projections with external instruments (LP-IV) following Stock and Watson (2018), which do not impose VAR dynamics.

4.2 Data and Variables

The VAR includes six variables:

$$\mathbf{Y}_t = \left(\text{SENT}_t \quad \text{UNC}_t \quad \text{ROP}_t \quad \text{WOP}_t \quad \text{WIP}_t \quad \text{CPI}_t\right)', \tag{6}$$

where SENT_t and UNC_t are the standardized aggregate narrative sentiment and uncertainty indices (constructed in Section 3.3), $\text{ROP}_t = 100 \times \log([\text{WTI}_t/\text{CPI}_t] * 100)$ is the real oil price, $\text{WOP}_t = 100 \times \log(\text{OILPRO}_t)$ is world oil production, $\text{WIP}_t = 100 \times \log(\text{INDPRO}_t)$ is world industrial production, and $\text{CPI}_t = 100 \times \log(\text{CPI}_t)$ is the consumer price index. For the uncertainty system, the ordering of the first two variables is reversed.

The real oil price is constructed using the WTI spot price from the U.S. Energy Information Administration (EIA), available from January 1986, deflated by the CPI. World industrial production (INDPRO) is from Baumeister and Hamilton (2019),and the consumer price index (CPI) is from the FRED-MD database (McCracken and Ng, 2016). World crude oil production is from the EIA. All variables enter in $100 \times \log$ levels except the text indices, which are standardized. The lag order is $p = 6$.

The estimation sample runs from 1987:02 to 2022:12 ($T = 419$ monthly observations after purging lags), determined by the intersection of EIA oil price availability (1986:01), text index coverage, and the loss of initial observations from instrument purging. Figure 5 displays the six VAR variables over the estimation sample.

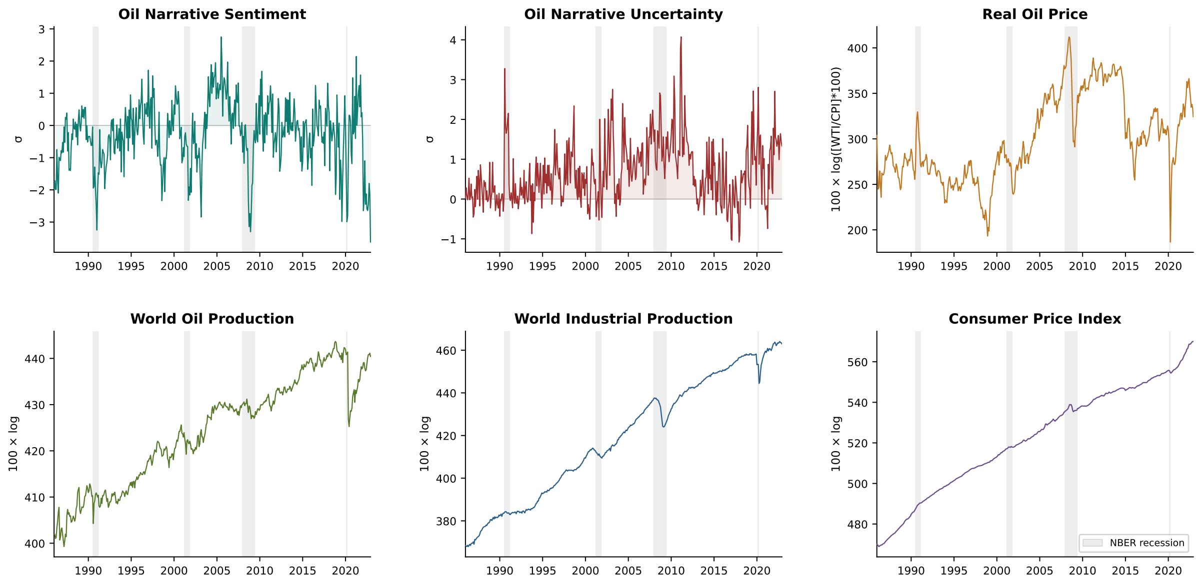


Figure 5: VAR variables, 1986:01–2022:12. Top row: oil narrative sentiment (standardized), oil narrative uncertainty (standardized), real oil price ($100 \times \log([\text{WTI}/\text{CPI}] \times 100)$). Bottom row: world crude oil production ($100 \times \log$), world industrial production ($100 \times \log$), consumer price index ($100 \times \log$). Shaded vertical bars denote NBER recession dates.

4.3 Instrument Construction

The identification strategy requires external instruments that are correlated with the aggregate narrative indices but capture variation from distinct, identifiable sources. We propose a novel approach: using the *within-corpus narrative type decomposition* as the source of instrumental variation.

The five type-specific sentiment indices (geopolitical, OPEC, speculation, supply disruption, and energy transition) each measure the directional tone of coverage from a different narrative source. By construction, the aggregate sentiment index is a weighted average of these type-specific components (plus general oil); the type-specific indices are therefore mechanically relevant for the aggregate. The key insight is that each type captures variation from a *different driver* of oil market news: geopolitical events operate through conflict and sanctions, OPEC decisions operate through cartel coordination, speculation operates through financial market positioning, supply disruptions operate through physical infrastructure, and energy transition operates through policy and technology. These are economically distinct sources of narrative variation, which is what makes them valid instruments because, they satisfy relevance by construction and exogeneity because they capture different *sources* of narrative information rather than the same shock measured with different noise.

This "narrative heterogeneity as identification" approach differs from existing strategies in the oil market literature. Känzig (2021) identifies oil supply news shocks using high-frequency price movements around OPEC announcements (a single-event), single-instrument strategy. Gavriilidis et al. (2026) construct a narrative instrument from the media impact of 146 classified climate policy events. Our approach replaces narrow event windows with the full richness of continuous narrative variation, decomposed by source, yielding five instruments per shock that jointly produce strong first stages. The approach is also portable: any text-based index that can be decomposed by source or topic can use the same strategy.

The five purged sentiment instruments form the instrument vector for the sentiment system:

$$\mathbf{Z}_t^{\text{sent}} = \left(\hat{\eta}_t^{\text{geo,sent}} \quad \hat{\eta}_t^{\text{opec,sent}} \quad \hat{\eta}_t^{\text{spec,sent}} \quad \hat{\eta}_t^{\text{supply,sent}} \quad \hat{\eta}_t^{\text{trans,sent}} \right)'. \tag{7}$$

Analogously, five purged uncertainty instruments form $\mathbf{Z}_t^{\text{unc}}$.

4.4 Instrument Purging

Following Romer and Romer (2004) and Miranda-Agrippino and Ricco (2021), each type-specific index is purged of predictable variation before use as an instrument. The purging regression for instrument j in dimension $d \in \{\text{sent}, \text{unc}\}$ is:

$$s_t^{j,d} = c + \sum_{\ell=1}^{12} \alpha_\ell s_{t-\ell}^{j,d} + \sum_{\ell=1}^{12} \boldsymbol{\beta}_\ell' \mathbf{x}_{t-\ell} + \eta_t^{j,d}, \tag{8}$$

where $\mathbf{x}_t = (\Delta \log \text{ROP}_t,\ \Delta \log \text{WOP}_t,\ \Delta \log \text{WIP}_t,\ \Delta \log \text{CPI}_t)'$ is the vector of macroeconomic state variables. The residual $\hat{\eta}_t^{j,d}$ replaces the raw index as the purged instrument.

This step is essential. It ensures that the instruments capture genuinely *new* narrative information (the surprise component of type-specific media coverage) rather than responding to lagged macroeconomic conditions. The purging is analogous to the high-frequency identification window in Känzig (2021): just as Känzig's OPEC announcement returns capture only the surprise component of OPEC decisions by using a narrow event window, the purging regression removes the predictable component of narrative variation over time.

4.5 Instrument Validation

We validate the purged instruments along four dimensions: Granger causality, serial correlation, first-stage strength, and over-identification test.

Granger causality. Valid external instruments must be unpredictable from lagged macroeconomic conditions; otherwise, they may capture systematic responses to past shocks rather than genuinely new narrative information. Table 4 reports F-statistics from regressions of each instrument on six lags of the four macroeconomic state variables. All ten purged instruments are unpredictable from lagged macroeconomic conditions, with p-values at or above 1.000 in every case. The joint Fisher tests confirm this at the system level: the combined p-value for all five sentiment instruments, all five uncertainty instruments, and all ten instruments jointly are each indistinguishable from unity. Prior to purging, three instruments exhibited significant macro forecastability: OPEC sentiment ($p < 0.001$), speculation sentiment ($p < 0.001$), and OPEC uncertainty ($p = 0.044$), validating the necessity of the purging step.

Table 4: Granger Causality Tests

	Raw (unpurged)		Purged	
Instrument	F-stat	p-value	F-stat	p-value
Sentiment instruments				
Geopolitical	0.96	0.526	0.00	1.000
OPEC	3.18	**0.000**	0.01	1.000
Speculation	4.23	**0.000**	0.00	1.000
Supply	0.92	0.571	0.02	1.000
Transition	0.97	0.511	0.01	1.000
Uncertainty instruments				
Geopolitical	0.82	0.715	0.03	1.000
OPEC	1.57	**0.044**	0.01	1.000
Speculation	0.98	0.491	0.01	1.000
Supply	1.07	0.372	0.04	1.000
Transition	1.11	0.333	0.22	1.000
Joint tests (Fisher χ^2)				
Sentiment (5 instruments)	74.21	**0.000**	0.00	1.000
Uncertainty (5 instruments)	12.50	0.253	0.00	1.000
All (10 instruments)	86.71	**0.000**	0.00	1.000

Notes: F-statistics from regressions of each instrument on 6 lags of $\Delta \log \mathrm{WIP}$, $\Delta \log \mathrm{CPI}$, $\Delta \log \mathrm{ROP}$, and $\Delta \log \mathrm{WOP}$. Bold p-values indicate rejection at the 5% level. Purging regresses each instrument on 12 own lags and 12 lags of the four state variables; the residual replaces the raw instrument. Joint tests use Fisher's method ($-2 \sum \log p_i \sim \chi^2(2k)$). Sample: 1987:02–2022:12.

Serial correlation. Valid instruments in the Proxy-SVAR framework should behave as innovation sequences, serially uncorrelated by construction if the purging successfully removes predictable variation. The Ljung-Box Q tests reported in Table 5 confirm that all ten purged instruments are serially uncorrelated at every tested lag length ($Q(6)$ through $Q(24)$, all $p > 0.37$). Prior to purging, nine of 10 raw instruments reject the null of no serial correlation at the 5% level for $Q(6)$, and 8 of 10 reject through $Q(24)$. The sole exception is transition uncertainty, which exhibits no significant autocorrelation even before purging—consistent with the sparse and episodic nature of energy transition coverage in the earlier part of the sample.

Table 5: Autocorrelation Diagnostics: Ljung-Box Q Tests

	$Q(6)$		$Q(12)$		$Q(18)$		$Q(24)$	
Instrument	Raw	Purged	Raw	Purged	Raw	Purged	Raw	Purged
Sentiment instruments								
Geopolitical	**0.000**	0.997	**0.000**	1.000	**0.000**	0.998	**0.000**	0.927
OPEC	**0.000**	0.999	**0.000**	1.000	**0.000**	0.792	**0.000**	0.730
Speculation	**0.001**	1.000	**0.003**	1.000	**0.004**	0.976	**0.000**	0.779
Supply	**0.004**	0.999	**0.005**	1.000	**0.027**	1.000	0.084	1.000
Transition	**0.023**	0.994	**0.028**	1.000	**0.005**	0.678	**0.019**	0.647
Uncertainty instruments								
Geopolitical	**0.012**	1.000	**0.024**	1.000	0.073	0.988	0.090	0.990
OPEC	**0.000**	1.000	**0.000**	1.000	**0.000**	1.000	**0.000**	1.000
Speculation	**0.000**	0.994	**0.000**	0.992	**0.000**	0.772	**0.000**	0.409
Supply	**0.000**	0.999	**0.000**	1.000	**0.002**	0.779	**0.001**	0.371
Transition	0.823	0.999	0.211	1.000	0.247	0.987	0.270	0.969

Notes: p-values from the Ljung-Box Q test for serial correlation at lags 6, 12, 18, and 24. "Raw" refers to the unpurged type-specific instruments; "Purged" refers to residuals from regressing each instrument on 12 own lags and 12 lags of the four macro state variables. Bold values indicate rejection at the 5% level. Sample: 1987:02–2022:12.

First-stage strength. Table 6 reports the first-stage regression results for both systems. For the sentiment shock, the joint F-statistic from the five purged type-specific instruments is 23.0 (HC1-robust: 21.1), well above the Stock and Yogo (2005) critical value of 16.38 for 5% maximal IV size distortion with five instruments. The R^2 is 21.5%, indicating that the type-specific instruments jointly explain roughly one-fifth of the variation in the reduced-form sentiment residual. Among individual instruments, speculation sentiment is the strongest ($F = 70.1$), followed by OPEC ($F = 25.4$), geopolitical ($F = 14.6$), and transition ($F = 11.3$). Supply sentiment is individually weak ($F = 4.4$) but contributes to the joint first stage.

For the uncertainty shock, the joint F-statistic is 31.6 (HC1-robust: 15.3), with $R^2 =$ 27.4%. The composition of instrument strength differs markedly: speculation ($F =$

66.4) and geopolitical uncertainty ($F = 53.4$) dominate, followed by supply ($F = 20.0$). OPEC ($F = 5.9$) and transition ($F = 3.6$) uncertainty are individually weak.

We supplement the conventional diagnostics with the effective F-statistic of Montiel Olea and Pflueger (2013), which accounts for heteroskedasticity and serial correlation in the first-stage regression via a Newey–West HAC covariance estimator. The effective F is 23.4 for the sentiment system and 13.5 for the uncertainty system. Both exceed the Montiel Olea and Pflueger (2013) critical value of 11.68 for at most 10% worst-case Nagar bias with five instruments; the sentiment system additionally clears the more demanding 5% threshold of 18.76. These results provide a stronger diagnostic than the conventional rule-of-thumb of $F > 10$, which assumes homoskedastic, serially uncorrelated errors.

The shift in instrument ranking across systems is economically informative. Sentiment is driven primarily by narratives about *what market participants think and do* (speculative positioning and OPEC decisions) consistent with a demand-side information channel. Uncertainty is driven by narratives about *sources of unpredictability* (geopolitical instability and speculative froth) consistent with a real options channel where producers respond to the *breadth* of possible outcomes rather than the *direction* of expectations.

Over-identification. Because the system is over-identified ($k = 5$ instruments for one endogenous variable, yielding four over-identifying restrictions), we can test whether all five instruments agree on the same structural shock using the Sargan J-test. Table 7 reports the results. The uncertainty system passes comfortably: no individual equation rejects at the 5% level, and the system-wide J-statistic has $p = 0.415$ (Panel B). The five uncertainty instruments are mutually consistent with a single structural shock.

The sentiment system shows more tension. The oil price and CPI equations reject at the 5% level ($p = 0.030$ and $p = 0.026$, respectively), while the remaining three equations pass ($p > 0.12$). The system-wide test rejects ($p = 0.010$). To identify the source of this tension, Panel C reports the C-statistic from sequentially dropping each instrument. The OPEC sentiment instrument is the sole source of rejection ($C = 9.90$, $p = 0.002$); all other instruments pass ($p > 0.08$). This is economically interpretable: OPEC narratives carry direct price information beyond their role as a proxy for aggregate sentiment—actual quota decisions and compliance signals move prices through channels other than narrative tone alone. Importantly, dropping the OPEC instrument leaves a joint $F = 23.8$ from the remaining four instruments, well above all weak-instrument thresholds, and the qualitative IRF results are unchanged.

Table 6: First-Stage Diagnostics

Panel A: Joint Instruments (5 instruments, 1 endogenous variable)

	Standard F	HC1-Robust F	Effective F	R^2	Adj. R^2	T
Sentiment	23.0	21.1	23.4	21.5%	20.6%	425
Uncertainty	31.6	15.3	13.5	27.4%	26.5%	425

Montiel Olea–Pflueger critical values ($k = 5$)	
$\tau = 5\%$ max bias	18.76
$\tau = 10\%$ max bias	11.68
$\tau = 20\%$ max bias	7.98

Panel B: Individual Instruments

	Sentiment			Uncertainty		
Instrument	Std. F	Eff. F	R^2	Std. F	Eff. F	R^2
Geopolitical	14.6	12.6	3.3%	53.4	22.0	11.2%
OPEC	25.4	22.7	5.7%	5.9	3.5	1.4%
Speculation	70.1	34.9	14.2%	66.4	10.4	13.6%
Supply	4.4	2.2	1.0%	20.0	11.1	4.5%
Transition	11.3	9.5	2.6%	3.6	3.0	0.8%

Montiel Olea–Pflueger critical values ($k = 1$)	
$\tau = 10\%$ max bias	23.11
$\tau = 20\%$ max bias	15.06

Notes: Panel A reports the standard, heteroskedasticity-robust (HC1), and effective F-statistics from the joint first-stage regression of the reduced-form VAR residual on all five purged type-specific instruments. The effective F-statistic of Montiel Olea and Pflueger (2013) uses a Newey–West HAC covariance estimator (5 lags) to account for both heteroskedasticity and serial correlation. Under the null of weak instruments, the effective F is compared to critical values $c_\alpha(\tau)$ from Montiel Olea and Pflueger (2013); τ denotes the worst-case Nagar bias as a fraction of the OLS bias. Both systems exceed the $\tau = 10\%$ threshold ($c = 11.68$), indicating that IV bias is at most 10% of OLS bias. Panel B reports individual (bivariate) first-stage results for each instrument separately, with $k = 1$ critical values. Sample: 1987:02–2022:12.

Table 7: Over-Identification Tests

Panel A: Narrative Sentiment Shock

Equation	J-statistic	p-value	
Other index	3.405	0.493	
Real oil price	10.710	0.030	$*$
World oil prod	5.229	0.265	
World IP	7.141	0.129	
CPI	11.083	0.026	$*$
System	37.568	0.010	$**$

Panel B: Narrative Uncertainty Shock

Equation	J-statistic	p-value
Other index	4.934	0.294
Real oil price	5.489	0.241
World oil prod	2.888	0.577
World IP	5.697	0.223
CPI	1.693	0.792
System	20.701	0.415

Panel C: Instrument Exclusion (C-test), Sentiment Shock

Dropped instrument	F (remaining 4)	C-statistic	p-value
Geopolitical	26.1	1.901	0.168
OPEC	23.8	9.898	0.002 $**$
Speculation	12.4	2.924	0.087
Supply	28.1	0.359	0.549
Transition	27.1	0.012	0.913

Notes: Panels A and B report Sargan J-statistics from 2SLS regressions of each reduced-form VAR residual $u_{j,t}$ on the five purged type-specific instruments, using the fitted value $\hat{u}_{1,t}$ from the first-stage regression. Under H_0 (all instruments valid), $J = T \times R^2 \sim \chi^2(4)$, where 4 is the number of over-identifying restrictions ($k - 1 = 5 - 1$). The system statistic sums the equation-level J-statistics; under H_0 it is distributed $\chi^2(20)$. Panel C reports the C-statistic (difference-in-J) for the sentiment system, testing the marginal validity of each instrument conditional on the remaining four being valid; $C \sim \chi^2(1)$ under H_0. $*$ $p < 0.05$; $**$ $p < 0.01$.

5 Empirical Results

5.1 Narrative Sentiment Shock

Figure 6 displays impulse responses to a one-standard-deviation *bullish* narrative sentiment shock, identified using the five purged type-specific sentiment instruments in the Proxy-SVAR framework. The first-stage F-statistic is 23.0 (robust: 21.1, effective: 23.4), with $R^2 = 21.5\%$.

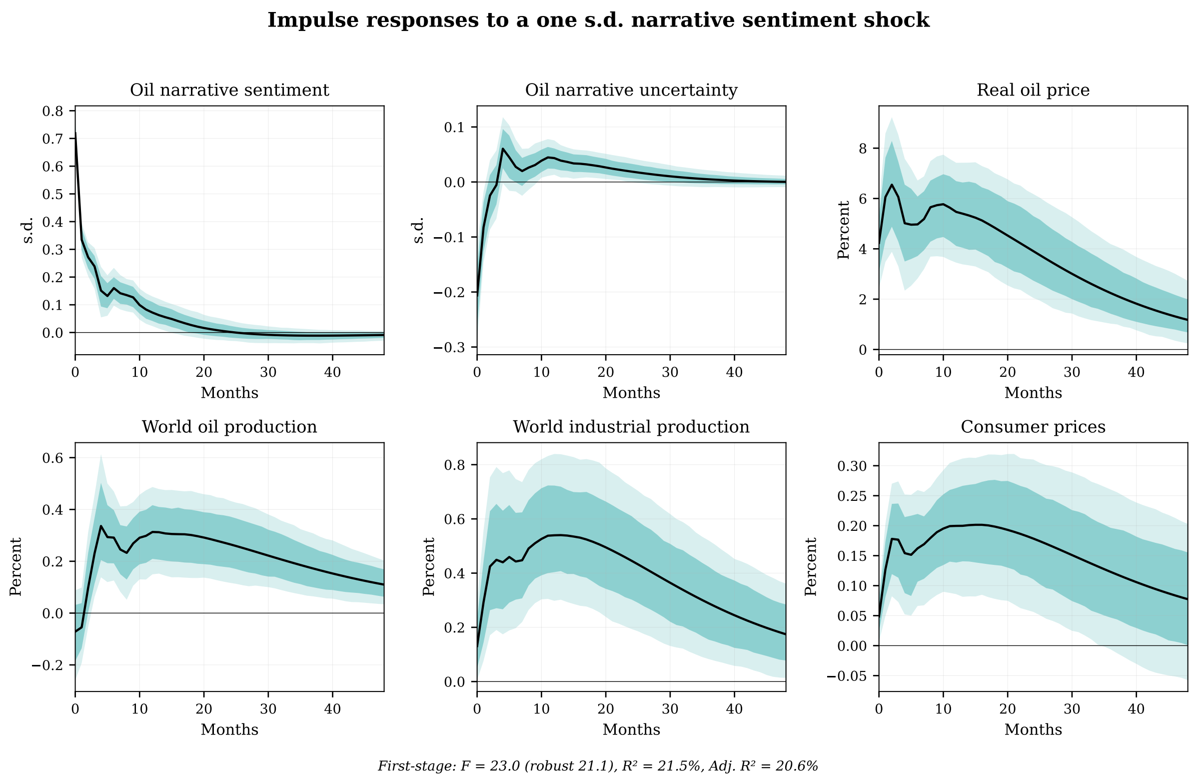


Figure 6: Impulse responses to a one standard deviation narrative sentiment shock. Dark shaded region: 68% MBB confidence bands. Light shaded region: 90% MBB confidence bands. Solid line: point estimate. The shock is identified using five purged type-specific sentiment instruments in the Proxy-SVAR framework. Sample: 1987:02–2022:12, VAR(6), 10,000 bootstrap replications.

Real oil price. The price increases by approximately 5% on impact and rises to a peak of roughly 6% around months 5–10, remaining significantly positive through 30 months before gradually reverting toward zero. This persistent price response, lasting over two years is consistent with the view that narratives shift beliefs about future fundamentals, generating persistent demand effects.

World industrial production. Output rises gradually to approximately 0.5% by months 10–15, significant at the 68% level through 30 months. The positive co-movement of prices and output is the defining signature of a demand-side shock: a supply-driven price increase would depress output, not raise it. This pattern rules out the interpretation that the sentiment shock captures supply-side news and instead points to the demand-side information channel developed in Section 2.

World oil production. Production rises modestly by 0.2–0.3% over months 5–15 and remains persistently positive at longer horizons. The positive production response reinforces the demand interpretation: producers respond to sustained price increases by expanding output, consistent with supply responding endogenously to a demand-driven price signal rather than being the source of the shock.

Consumer prices. The CPI increases gradually to approximately 0.2% by month 10, with effects that persist through the 48-month horizon, significant at 68% throughout. This pass-through from narrative sentiment to consumer prices operates via the oil price channel and has direct implications for monetary policy: shifts in the narrative environment of oil markets generate inflationary pressure that central banks must accommodate or offset.

Cross-effect on uncertainty. There is no significant effect of the sentiment shock on narrative uncertainty. The point estimate shows a brief negative dip of approximately -0.2 standard deviations on impact that reverts to zero within 5 months, with the 68% bands encompassing zero at all but the shortest horizons. This confirms that the sentiment and uncertainty dimensions capture distinct structural shocks rather than reflecting common variation.

The impulse response pattern; persistent price increase, positive output co-movement, gradual CPI pass-through, expanding production, and no cross-contamination of uncertainty is the textbook signature of a demand-side information shock. Bullish narrative sentiment shifts market expectations about future oil market conditions, raises willingness to hold inventories, and bids up current prices, with real activity responding positively through the expenditure channel. These results are consistent with the demand-side information channel hypothesized in Section 2 and with the finding in Abeyie and Hanson (2025) that oil market narratives embed demand information.

5.2 Narrative Uncertainty Shock

Figure 7 displays impulse responses to a one-standard-deviation narrative uncertainty shock, identified using the five purged type-specific uncertainty instruments. The first-stage F-statistic is 31.6 (robust: 15.3, effective: 13.5), with $R^2 = 27.4\%$.

The uncertainty shock generates a qualitatively distinct pattern from sentiment, consistent with the real options channel developed in Section 2.

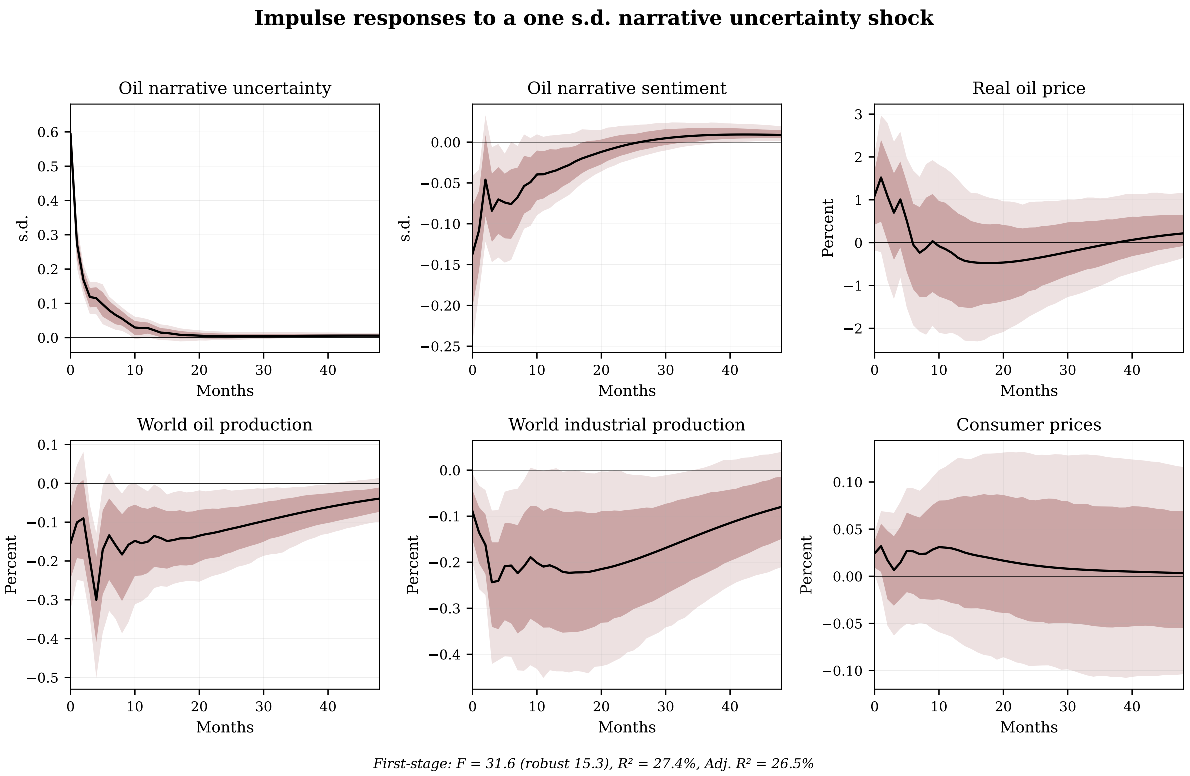


Figure 7: Impulse responses to a one standard deviation narrative uncertainty shock. Dark shaded region: 68% MBB confidence bands. Light shaded region: 90% MBB confidence bands. Solid line: point estimate. The shock is identified using five purged type-specific uncertainty instruments in the Proxy-SVAR framework. Sample: 1987:02–2022:12, VAR(6), 10,000 bootstrap replications.

World oil production. Production declines by approximately 0.3% on impact and remains depressed at 0.10–0.20% through the full 48-month horizon, significant at the 68% level. This persistent output contraction is the defining feature of the uncertainty shock and is consistent with real options theory: when ambiguity about future oil market conditions rises, producers delay irreversible drilling and investment decisions in a wait-and-see response (Bernanke, 1983; Bloom, 2009; Kellogg, 2014). The persistence of the effect; production remains depressed for four years, reflects the long lead times and capital intensity of upstream oil investment.

World industrial production. Output declines gradually to approximately 0.2–0.3% below baseline by months 10–20, consistent with uncertainty spillovers from the energy sector to broader economic activity. The response is significant at the 68% level through 30 months. This transmission from oil market narrative uncertainty to global real activity operates through both the direct investment channel (reduced energy-sector capital expenditure) and the indirect confidence channel (heightened ambiguity dampening non-energy investment and consumption decisions).[9]

Real oil price. The response of the real oil price is muted, somewhat non-monotonic, and statistically weak. The point estimate shows a brief positive movement of roughly 0.5–1% in the first few months, followed by a decline toward zero around months 10–15 and a gradual recovery thereafter. However, the response is not persistently significant at the 90% level at any horizon. This pattern is consistent with offsetting forces. On the one hand, higher narrative uncertainty depresses real activity and weakens oil demand, placing downward pressure on prices. On the other hand, uncertainty also restrains oil production through a wait-and-see or real-options channel, which puts upward pressure on prices. The resulting net effect is therefore ambiguous, in contrast to the sentiment channel, where the price response is more clearly directional.

Consumer prices. The CPI response is small and statistically insignificant, with point estimates fluctuating between 0 and 0.05% across the 48-month horizon. The absence of a meaningful CPI response reflects the offsetting forces described above: the demand-depressing effect of uncertainty and the supply-reducing effect generate opposing pressures on the general price level that largely cancel.

Cross-effect on sentiment. Unlike the sentiment shock, which produces no significant effect on uncertainty, the uncertainty shock generates a persistent negative effect on narrative sentiment of approximately 0.10–0.15 standard deviations, significant at 68% through 25 months. This asymmetry is interpretable: sustained ambiguity in oil market coverage gradually erodes bullish tone, as media narratives shift from directional reporting to hedged, uncertain framing. The effect is modest in magnitude, confirming that the two narrative dimensions capture primarily distinct variation, but the asymmetry; uncertainty depresses sentiment, but sentiment does not raise uncertainty, is informative about the narrative generating process.

The impulse response pattern; persistent production decline, reduced global output, modest upward pressure on oil price effects, negligible CPI response, and gradual erosion of sentiment, is the signature of a supply-side real options shock. Elevated narrative uncertainty causes producers to delay irreversible investment, suppressing oil production for years without generating the decisive price and output co-movement that characterizes demand-driven shocks. These results confirm the two-channel hypothesis: narrative sentiment operates through demand-side information (prices and output move together), while narrative uncertainty operates through supply-side real

[9] This is consistent with the broader literature on uncertainty and real activity. Jurado et al. (2015) show that macroeconomic uncertainty shocks depress output persistently, and Baker et al. (2016) document that policy uncertainty is associated with reduced investment and hiring. The narrative uncertainty index proposed here captures a domain-specific form of such ambiguity.

options (production contracts with ambiguous price effects). The two shocks, extracted from the same newspaper corpus, transmit to the macroeconomy through fundamentally different mechanisms.

5.3 The Real Options Channel: Evidence from Drilling Activity

The conceptual framework in Section 2 suggests that narrative uncertainty may transmit to the macroeconomy through a supply-side delay channel: heightened ambiguity raises the option value of waiting and leads firms to postpone irreversible drilling and investment decisions (Bernanke, 1983; Bloom, 2009; Kellogg, 2014). To assess this interpretation more directly, I augment the baseline VAR with the Baker Hughes U.S. oil-directed rig count following the marginal VAR approach of Gertler and Karadi (2015). Rig activity is a useful margin for this exercise because it lies closer to drilling and investment decisions than aggregate crude oil production, which adjusts more slowly and may reflect a wider set of contemporaneous forces.

Figure 8 displays the rig-count response to each narrative shock. The contrast is sharp. A one-standard-deviation narrative uncertainty shock produces a persistent decline in rig activity. The response is close to zero on impact, but turns increasingly negative over the following months, reaching approximately −1.8% after six months, −2.7% after twelve months, and a trough of roughly −2.8% around month 9. The delayed and persistent nature of the decline is economically intuitive: firms do not instantly shut down drilling, but as uncertainty remains elevated they scale back new drilling activity, consistent with a wait-and-see response.

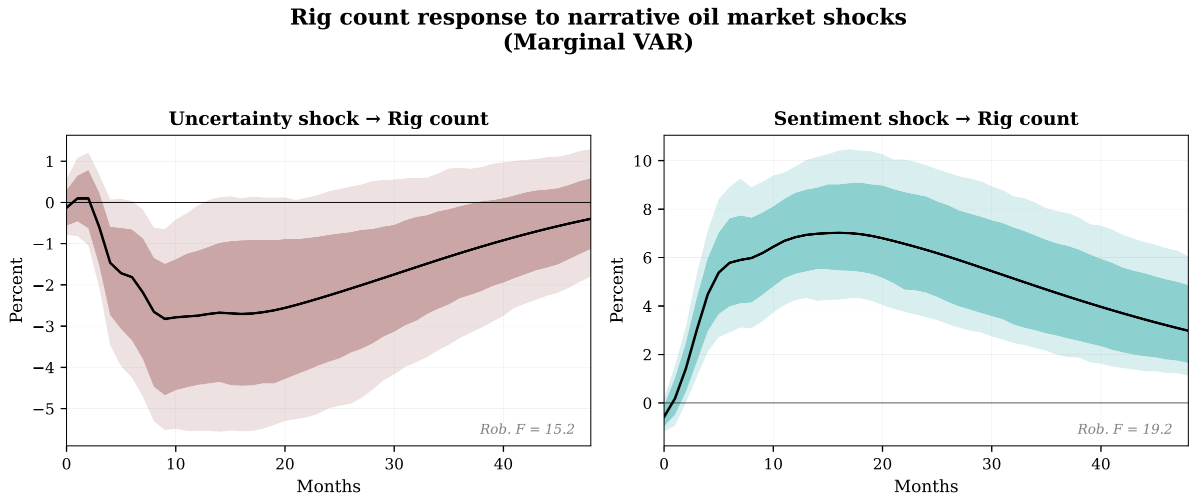


Figure 8: Rig count response to narrative oil market shocks (marginal VAR). Left panel: response to a one standard deviation narrative uncertainty shock. Right panel: response to a one standard deviation narrative sentiment shock. Dark shaded region: 68% MBB confidence bands. Light shaded region: 90% MBB confidence bands. Sample: 1987:08–2022:12

The sentiment shock produces the opposite pattern. Following a small impact response, rig counts rise strongly and persistently, increasing by approximately 5.8% after six months, 6.8% after twelve months, and peaking at about 7.0% around month 16. This response is consistent with the demand-side information channel developed earlier: bullish oil-market narratives strengthen expectations about future market conditions, raise the expected return to drilling, and induce an endogenous expansion in rig deployment.

The rig-count responses provide direct evidence that the two narrative dimensions affect the drilling margin in markedly different ways. Narrative uncertainty suppresses rig activity, while narrative sentiment stimulates it. This contrast strengthens the broader interpretation of the paper's main results. The decline in world oil production following an uncertainty shock is not merely an aggregate quantity response; it is accompanied by a reduction in drilling activity itself, consistent with a supply-side delay channel. Likewise, the positive production response to sentiment is mirrored by a substantial increase in rig deployment, consistent with stronger expected returns and an endogenous supply response to improved market expectations.

5.4 Combined Shocks and the Stagflationary Scenario

The two-channel framework also generates a prediction for episodes in which bullish sentiment and elevated uncertainty coincide. Because the VAR is linear, the combined effect of simultaneous shocks equals the sum of their individual impulse responses. Figure 9 displays the resulting responses.

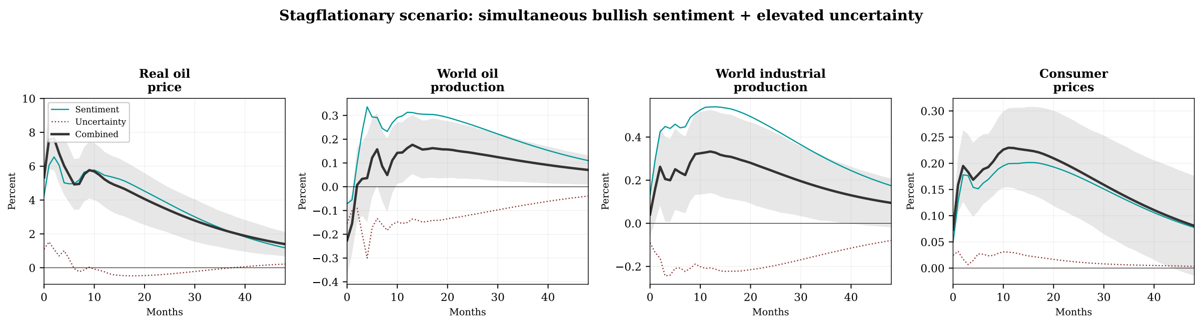


Figure 9: Simultaneous narrative sentiment and uncertainty shocks. Teal line: sentiment shock alone. Dark red dotted line: uncertainty shock alone. Black line: combined effect (sum of individual IRFs). Shaded region: 68% confidence band for the combined effect, computed by adding individual standard errors in quadrature. The combined shock produces a stagflationary pattern: oil prices rise strongly (+7.6%), oil production remains weak (−0.2%), and consumer prices increase (+0.2%).

The combined shock produces a distinctive macroeconomic pattern that neither shock generates on its own. Real oil prices rise strongly (+7.6%), as bullish sentiment strengthens demand expectations while elevated uncertainty restrains supply adjustment. Oil production, however, remains weak (-0.2%): the sentiment shock alone would raise production by about +0.3% as producers respond to stronger price incentives, but the uncertainty shock reduces production by roughly -0.3% through delayed drilling and investment, leaving little net increase in output. World industrial production rises only modestly (+0.3%), far less than under a pure sentiment shock, because the uncertainty-induced contraction partially offsets the demand-driven expansion. Consumer prices increase (+0.2%), driven primarily by oil-price pass-through from the sentiment component.

This combination of sharply higher oil prices, weak production, subdued activity, and rising consumer prices is a stagflationary signature. The key point is that neither narrative dimension alone generates this pattern. A sentiment shock alone looks like a demand-side information disturbance, with prices and activity moving together. An uncertainty shock alone looks like a supply-side delay disturbance, with activity and production falling but prices responding only weakly. Only when the two shocks occur together does the economy display the combination of rising prices and weak real activity that characterizes stagflationary oil episodes.

This perspective helps interpret major oil-market episodes such as the Gulf War (1990–91), the Arab Spring (2011), and the Russia-Ukraine conflict (2022), in which oil-market coverage likely became both more bullish and more uncertain at the same time. The framework therefore offers a decomposition that is not available in the standard three-shock model of Kilian (2009), which does not separately identify the directional and ambiguity components of oil-market news.

The stagflationary scenario does not affect all economies uniformly. To examine cross-country heterogeneity, I augment the baseline VAR with industrial production from six European economies using the marginal VAR approach. Figure 10 shows substantial variation across countries. Norway, the only oil producer in the panel, experiences the largest contraction (-0.77%), as the sentiment component depresses manufacturing through higher input costs (-0.65%) while the uncertainty component delays domestic drilling and investment (-0.54%). France also contracts (-0.32%), as the uncertainty drag dominates the modest demand boost from sentiment. By contrast, Italy ($+0.47\%$) and Spain ($+0.30\%$) experience positive net effects because the demand-side information channel more than offsets the uncertainty channel. Germany is close to stagnation ($+0.31\%$), while the Netherlands records the strongest positive response ($+0.53\%$). These results suggest that the effects of simultaneous sentiment and uncertainty shocks depend importantly on an economy's position in the oil supply chain and are not visible in aggregate analysis alone.

Heterogeneous European industrial production responses

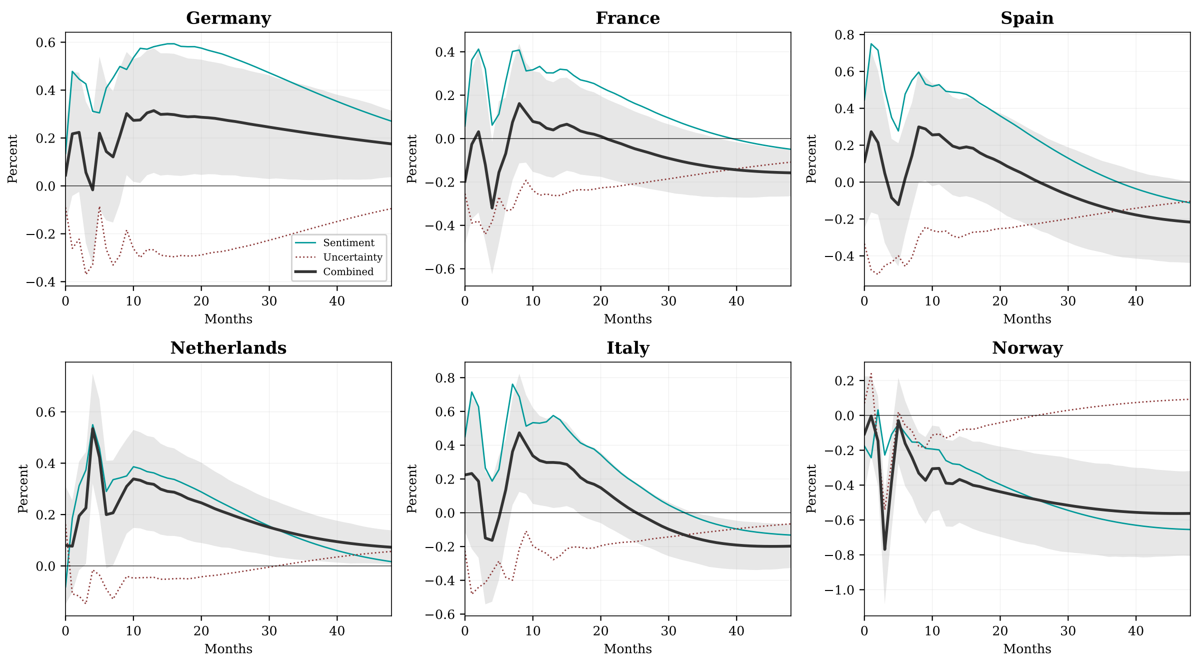
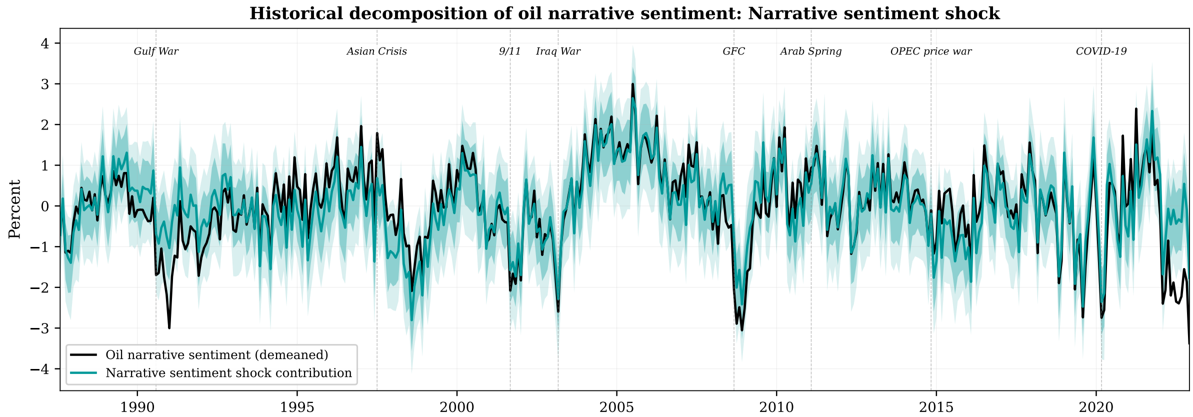


Figure 10: Combined stagflationary effect on European industrial production (marginal VAR). Each panel augments the baseline VAR with one country's industrial production. Teal line: sentiment shock alone. Dark red dotted line: uncertainty shock alone. Black line: combined effect (sum of individual IRFs). Shaded region: 68% confidence band for the combined effect. Norway, the sole oil producer, experiences the largest contraction (−0.77%). Italy and Spain are the most resilient as the demand channel dominates. Sample: 1987:02–2022:12, VAR(6).

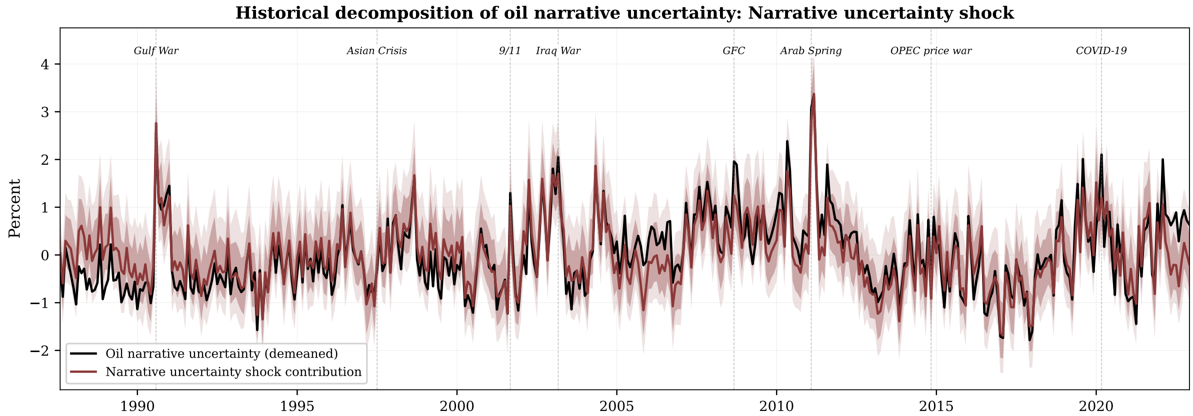
5.5 Historical Decomposition

Figure 11 displays the historical contribution of each identified narrative shock to its own instrumented variable; the aggregate sentiment and uncertainty indices, respectively. These decompositions serve as an event validation exercise: if the identified shocks are economically meaningful, their historical contributions should align with known oil market episodes.

Panel (a) shows the contribution of the narrative sentiment shock to the aggregate oil narrative sentiment index. The shock contribution tracks the actual index closely, capturing the major swings in oil market tone over the past three decades: the bearish turn during the Gulf War (1990), the deep pessimism of the Asian Financial Crisis (1997–98), the sustained bullish buildup during the Iraq War speculation and financialization wave (2002–07), the sharp bearish collapse during the Global Financial Crisis (2008), the prolonged pessimism during the OPEC price war (2014–16), and the extreme negative sentiment shock of COVID-19 (2020). The tight tracking between the shock contribution and the actual index (with the 68% confidence bands encompassing the realized series at most horizons) confirms that the Proxy-SVAR successfully isolates the structural component of narrative sentiment variation.



(a) Narrative sentiment shock



(b) Narrative uncertainty shock

Figure 11: Historical decomposition of oil narrative indices. Solid black line: demeaned aggregate index. Colored line: estimated contribution of the identified structural shock. Shaded regions: 68% and 90% MBB confidence bands. Panel (a): contribution of the narrative sentiment shock to the aggregate oil narrative sentiment index. Panel (b): contribution of the narrative uncertainty shock to the aggregate oil narrative uncertainty index. Vertical dashed lines mark major oil market events. Sample: 1987:02–2022:12, VAR(6), 10,000 bootstrap replications.

Panel (b) shows the contribution of the narrative uncertainty shock to the aggregate oil narrative uncertainty index. The shock contribution captures the episodic, spike-driven nature of oil market uncertainty: the Gulf War (1990), 9/11 (2001), the Arab Spring (2011, the largest identified uncertainty shock in the sample), and COVID-19 (2020). Between these episodes, the uncertainty shock contribution is small, reflecting the fact that baseline uncertainty is driven by slow-moving structural factors (e.g., market maturity, financialization) rather than by discrete narrative shocks. The concentration of large uncertainty shocks around geopolitical and crisis events is consistent with the first-stage results, where geopolitical and speculation uncertainty are the dominant instruments.

5.6 Forecast Error Variance Decomposition

Figure 12 displays the share of forecast error variance attributable to each narrative shock across all six VAR variables. The results quantify the relative importance of the two channels.

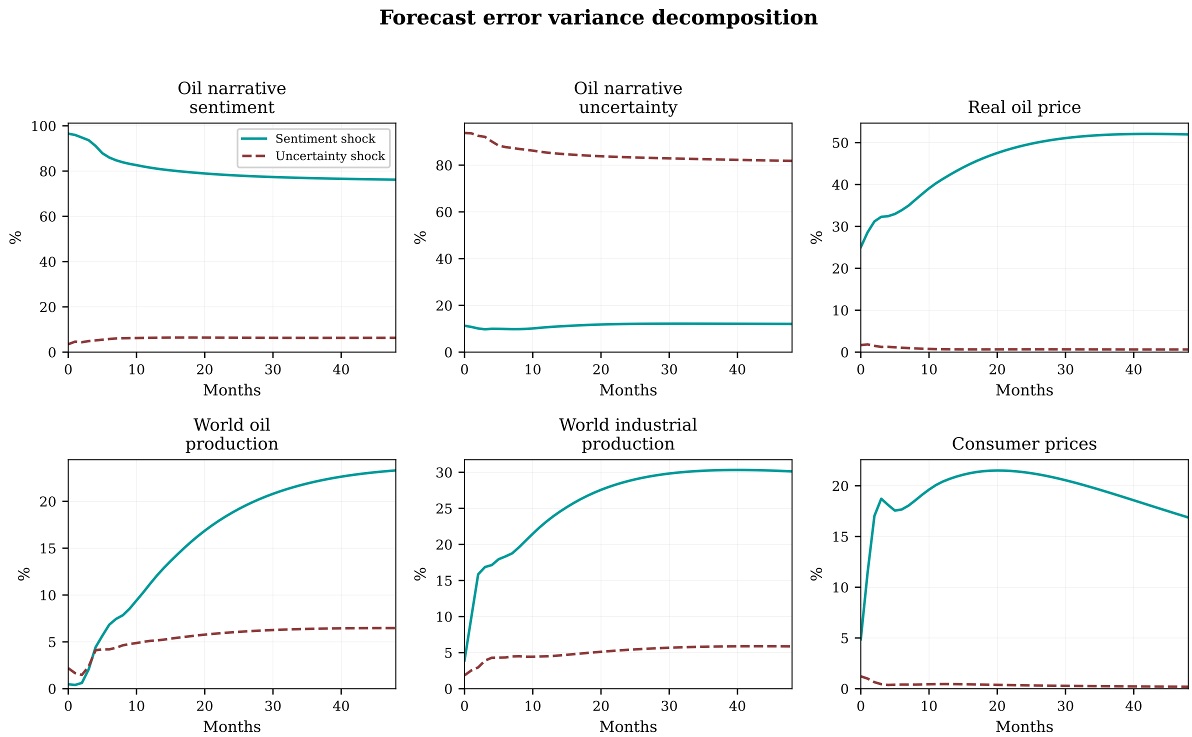


Figure 12: Forecast error variance decomposition. Each panel shows the share of forecast error variance attributable to the narrative sentiment shock (solid teal) and the narrative uncertainty shock (dashed dark red) at horizons 0–48 months. The sentiment and uncertainty shocks are identified from separate Proxy-SVAR systems. Sample: 1987:02–2022:12, VAR(6), 10,000 MBB replications.

The sentiment shock dominates the variance of oil prices and macroeconomic aggregates. It accounts for over 50% of real oil price variance at business-cycle horizons, rising from roughly 30% on impact to over 50% by month 24. It also explains approximately 30% of world industrial production variance and 20% of consumer price variance at the two-year horizon. By contrast, the uncertainty shock contributes negligibly to these variables—never exceeding 5% for oil prices, world IP, or CPI at any horizon.

The pattern reverses for world oil production. The uncertainty shock's contribution rises steadily from roughly 2% on impact to approximately 7% by month 48, while the sentiment shock's contribution grows more steeply to roughly 25% at longer horizons. The sentiment shock's production contribution reflects the endogenous supply response to demand-driven price increases; producers expand output when prices rise persistently, rather than a direct supply-side mechanism.

Each shock overwhelmingly explains its own index: the sentiment shock accounts for 75–95% of sentiment variance and the uncertainty shock accounts for 80–90% of uncertainty variance, with minimal cross-contamination (below 10% in both directions). This confirms that the two identified shocks capture distinct dimensions of narrative variation rather than reflecting common underlying factors.

Overall, the FEVD results reinforce the two-channel interpretation. Narrative sentiment is the dominant force behind oil price fluctuations and their macroeconomic transmission; operating through the demand-side information channel. Narrative uncertainty operates primarily through the supply-side real options channel, with its effects concentrated in production decisions.

6 Robustness

6.1 Placebo Tests

To confirm that the first-stage results reflect genuine narrative content rather than spurious correlation, I conduct two placebo exercises. Figure 13 reports the results.

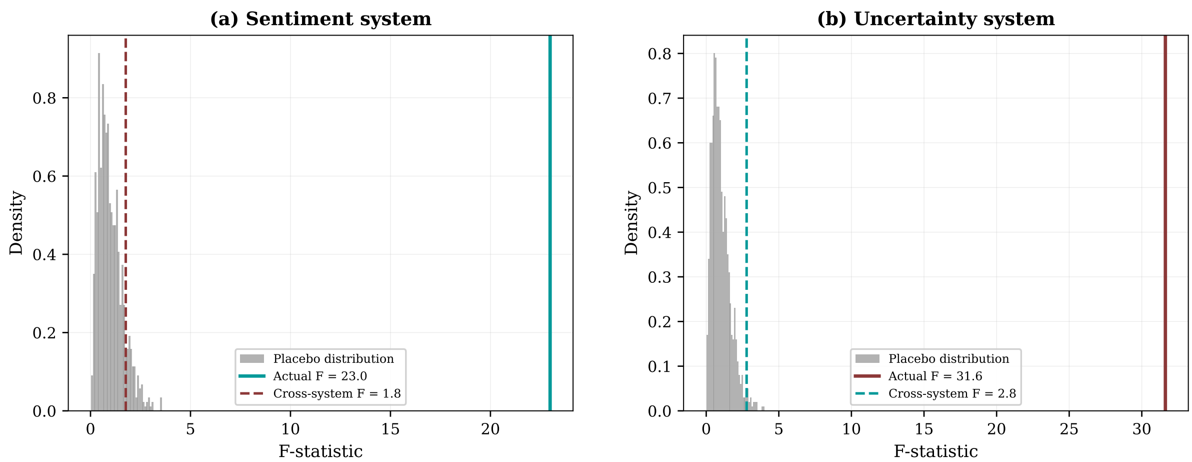


Figure 13: Placebo tests: permutation and cross-system instrument swap. Grey histograms: distribution of first-stage F-statistics from 1,000 random permutations of the instruments. Solid vertical line: actual F-statistic from the correctly assigned instruments. Dashed vertical line: F-statistic from swapping the instrument sets across systems (uncertainty instruments applied to the sentiment system, and vice versa). Sample: 1987:02–2022:12.

First, I randomly permute each instrument across time 1,000 times, destroying the temporal relationship with the VAR residuals while preserving the marginal distribution. The grey histograms in Figure 13 show the resulting placebo F-statistics, which are centered near unity in both systems. None of the 1,000 permutations produces an F-statistic exceeding the actual values of 23.0 (sentiment) or 31.6 (uncertainty). The implied permutation p-value is less than 0.001 in both cases.

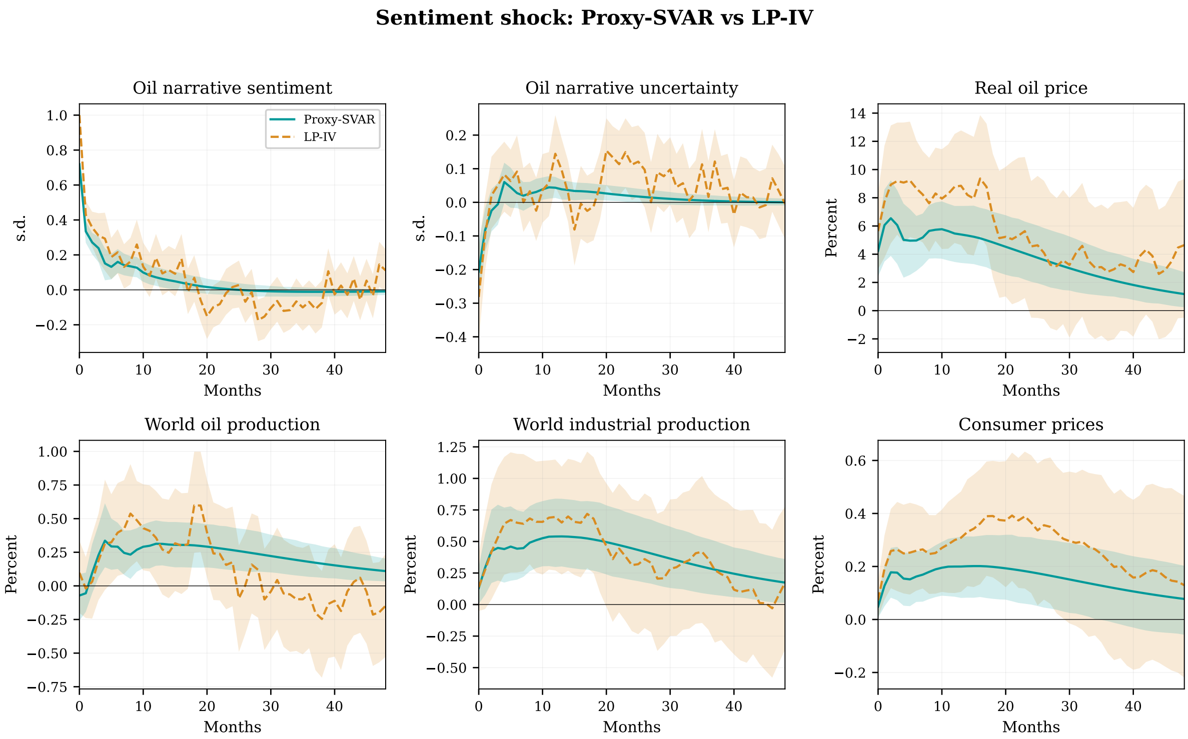
Second, I swap the instrument sets across systems, using the five uncertainty instruments to instrument sentiment and vice versa. The cross-system F-statistics are 1.8 (uncertainty instruments applied to sentiment) and 2.8 (sentiment instruments applied to uncertainty). Both fall within the placebo distribution and are indistinguishable from random noise. This confirms that the type-specific indices are shock-specific. Sentiment instruments predict sentiment but not uncertainty. Uncertainty instruments predict uncertainty but not sentiment. The two identified shocks draw on genuinely distinct dimensions of narrative variation rather than on a common oil news factor.

6.2 LP-IV vs. Proxy-SVAR

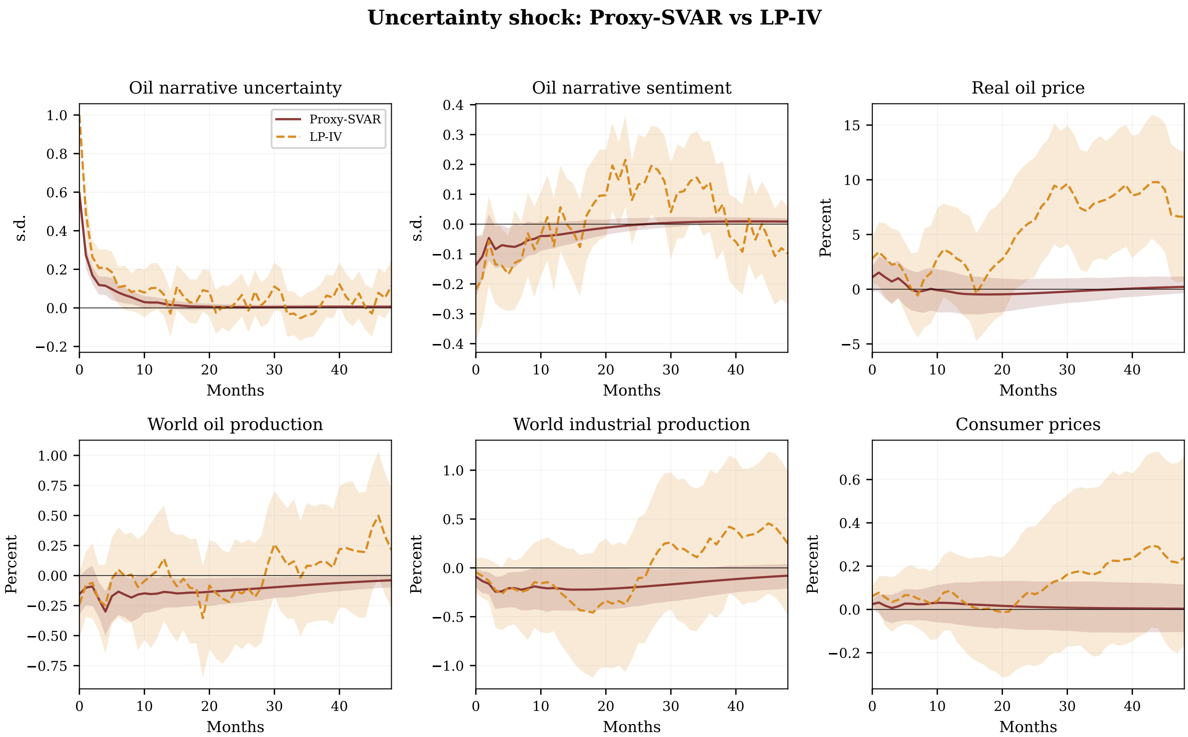
Figure 14 compares the Proxy-SVAR impulse responses with those obtained from local projections with external instruments (LP-IV), following Stock and Watson (2018). The LP-IV specification uses 1 lag and the same purged instruments, but does not impose VAR dynamics on the impulse response path.

Panel (a) reports the sentiment shock. The LP-IV point estimates (dashed orange) track the Proxy-SVAR (solid teal) closely for the key variables: the real oil price response peaks near 8–10% under LP-IV compared to 5–6% under the Proxy-SVAR, world industrial production follows a similar hump-shaped path, and the CPI pass-through is comparable in magnitude and persistence. LP-IV confidence bands are wider, as expected given the loss of efficiency from not imposing VAR dynamics. The qualitative conclusions are unchanged: bullish sentiment raises prices, output, production, and consumer prices.

Panel (b) reports the uncertainty shock. The Proxy-SVAR and LP-IV point estimates again agree on the sign and persistence of the key responses: world oil production declines, world industrial production falls, and the price response is ambiguous. The LP-IV estimates are noisier at longer horizons, particularly for the cross-effect on sentiment and for oil prices, which is expected given the model-free nature of local projections over a 48-month horizon. The core finding, that uncertainty suppresses quantities with modest increase in prices is robust to the choice of estimation method.



(a) Narrative sentiment shock



(b) Narrative uncertainty shock

Figure 14: Proxy-SVAR vs. LP-IV impulse responses. Solid colored line: Proxy-SVAR point estimate with 90% MBB confidence bands (shaded). Dashed orange: LP-IV point estimate with 90% MBB confidence bands (shaded). Sample: 1987:02–2022:12.

6.3 Excluding the OPEC Instrument

The C-test in Table 7 identified the OPEC sentiment instrument as the sole source of tension in the over-identification test ($C = 9.90$, $p = 0.002$). To verify that this does not affect the paper's conclusions, we re-estimate the sentiment Proxy-SVAR using only four instruments (geopolitical, speculation, supply, and transition), dropping OPEC sentiment entirely.

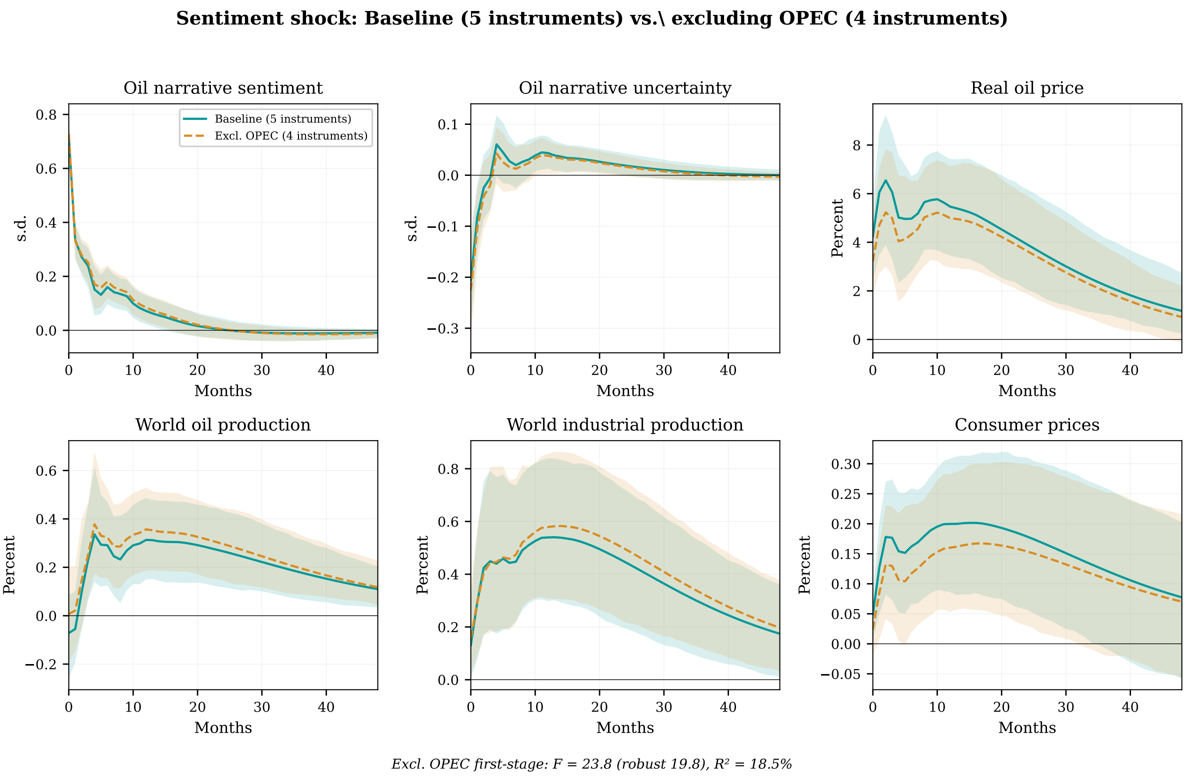


Figure 15: Robustness: excluding the OPEC sentiment instrument. Solid teal: baseline Proxy-SVAR with five instruments (90% confidence bands shaded). Dashed orange: Proxy-SVAR with four instruments (OPEC excluded). The four-instrument first stage remains strong ($F = 23.8$, robust $F = 19.8$, $R^2 = 18.5\%$). Sample: 1987:02–2022:12, VAR(6), 10,000 MBB replications.

Figure 15 compares the baseline five-instrument IRFs with the four-instrument specification. The point estimates are nearly identical across all six variables: the oil price peaks at approximately the same magnitude, world industrial production follows the same hump-shaped path, and the CPI pass-through is unchanged. The four-instrument first stage remains strong ($F = 23.8$), well above all weak-instrument thresholds. The marginal rejection in the over-identification test reflects the independent price information content of OPEC narratives, not a failure of the identification strategy.

6.4 Additional Robustness Checks

We conduct four further specification checks; full results are reported in Appendix C.

Alternative lag order. The baseline uses $p = 6$. Re-estimating with $p = 3$ and $p = 12$ produces qualitatively identical impulse responses for both shocks, with minor differences in confidence band width at longer horizons (Appendix Figures 20 and 21).

Leave-one-out instruments. Re-estimating the sentiment Proxy-SVAR five times, each time dropping one of the five instruments, produces point estimates that remain close to the baseline and within the 90% confidence bands in every case. All five leave-one-out specifications maintain joint F-statistics above 10, ranging from 12.4 (dropping speculation) to 28.1 (dropping supply). No single instrument drives the results (Appendix Figure 22).

Sub-sample stability. Splitting the sample at December 1999 yields consistent results across subsamples. The sentiment shock is well-identified in both halves (Robust $F = 12.1$ pre-2000, 15.5 post-2000) and drives oil prices in both sub-periods, with a larger magnitude in the post-2000 sample (+8.6% vs. +2.7%), consistent with financialization amplifying the demand-side information channel. The uncertainty shock is well-identified only in the post-2000 sample (Robust $F = 12.6$), where the production decline and activity contraction are larger than in the full sample (Appendix Figures 23 and 24; Table 9).

Cholesky identification. As a benchmark, we estimate the VAR with recursive (Cholesky) identification using the ordering $(\text{SENT}, \text{UNC}, \text{ROP}, \text{WOP}, \text{WIP}, \text{CPI})$. The Cholesky impulse responses are qualitatively similar to the Proxy-SVAR results for the sentiment shock but less precisely estimated for the uncertainty shock, where the recursive ordering assumption is harder to justify. The Proxy-SVAR is preferred because it does not require an ordering assumption and leverages the external instrument variation for identification (Appendix Figure 25).

7 Conclusion

This paper asks whether oil-market narratives have macroeconomic effects, and whether those effects differ depending on whether narratives convey directional information or ambiguity. The answer to both questions is yes.

The main finding is that bullish narrative sentiment and narrative uncertainty behave like different types of macroeconomic disturbances. A bullish narrative sentiment shock raises the real price of oil, increases world industrial production, raises world oil production, and gradually passes through to consumer prices. This pattern is consistent with a demand-side information channel in which bullish oil-market narratives shift expectations about future scarcity or future demand. By contrast, a narrative uncertainty shock reduces oil production and real activity while producing only a muted and weakly estimated response of the real oil price and consumer prices. This pattern is consistent with a supply-side delay channel in which heightened ambiguity raises the option value of waiting and slows the adjustment of partially irreversible production and investment decisions.

These results suggest that oil-market narratives contain macroeconomically relevant variation beyond what is captured by standard physical oil shocks alone. More broadly, they imply that existing oil-shock frameworks may miss an important distinction between oil-market news that shifts directional expectations and oil-market news that primarily raises ambiguity. Distinguishing between these two forms of narrative variation helps explain why some episodes of intense oil-market coverage generate price-output co-movement, while others primarily suppress production and activity without producing a comparably strong price response.

These results also imply that episodes in which narrative sentiment and narrative uncertainty rise together can generate a macroeconomic pattern that neither shock produces in isolation. When bullish oil-market narratives coincide with elevated ambiguity, stronger demand expectations push oil prices upward while uncertainty slows the adjustment of production and investment. The result is a stagflationary combination of rising oil prices, weak production, softer real activity, and higher inflation. This perspective helps interpret episodes of intense oil-market coverage in which prices surge but production and activity respond only weakly.

The broader implication is that narratives matter not only because they summarize oil-market events, but because they shape how those events are processed by firms, investors, and policymakers. Directional tone and ambiguity are not the same object, and they do not have the same macroeconomic consequences. Recognizing that distinction helps clarify how oil news transmits to the broader economy.

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Online Appendix

Narrative Shocks in the Oil Market: Evidence from 70 Years of News Coverage

April 9, 2026

A LLM Validation of Dictionary Classification

To assess the accuracy of the dictionary-based classification, I compare it against independent classifications from a large language model (Claude Sonnet 4) following the methodology of Gavriilidis et al. (2026).

A.1 Oil-Relevance Filter

Panel A of Table 8 reports the oil-relevance validation. A stratified random sample of 500 articles (250 dictionary-flagged as oil-relevant, 250 not) is independently classified by the LLM. Each article's text is truncated to the first 1,500 words. The dictionary achieves 86% accuracy, 80% precision, and 92% recall against the LLM benchmark, with a false positive rate of 20%. These results are consistent with rates typically observed for text-based indices in this literature (Baker et al., 2016; Caldara and Iacoviello, 2022; Gavriilidis et al., 2026).

The high recall (92%) confirms that the dictionary rarely misses articles the LLM considers oil-relevant. The 20% false positive rate reflects the dictionary's broader net: it flags articles that mention oil terms anywhere in the full text, while the LLM evaluates a truncated version. For the paper's purposes, high recall is more important than high precision, since type-specific sentiment and uncertainty scores are averaged across all flagged articles within each month, and occasional marginal inclusions are diluted by the large volume of clearly oil-relevant coverage.

A.2 Narrative Type Classification

Panel B of Table 8 reports the narrative type validation using binary LLM prompts. For each of 500 oil-relevant articles (stratified at approximately 80 per specific type), the LLM is asked five separate binary questions corresponding to the five narrative categories. This follows the approach of Gavriilidis et al. (2026), who use binary prompts to validate their climate policy uncertainty dictionary.

Four of five types achieve alignment rates consistent with published benchmarks. Energy transition has the highest alignment (96%), followed by OPEC (80%), geopolitical (75%), and supply disruption (70%). Speculation has a lower alignment rate (46%), which reflects the inherent overlap between financial market coverage and the underlying events that drive speculation. Articles about oil futures often discuss the geopolitical or supply events motivating financial positioning, and the LLM identifies the underlying driver rather than the financial framing.

Figure 16 displays the cross-contamination matrix. The off-diagonal patterns are economically interpretable: OPEC articles are frequently also flagged as geopolitical (49%), geopolitical articles as supply-related (26%), and supply articles as geopolitical (34%). These overlaps reflect the multi-dimensional nature of oil market narratives rather than classification errors. An OPEC production cut announced during a geopolitical crisis contains both OPEC coordination and geopolitical content. The dictionary assigns a single dominant type based on relative keyword counts; the LLM detects all relevant dimensions through contextual reading.

The key takeaway is that the dictionary approach produces classifications broadly consistent with a contextual language model for the categories that matter most for identification. The type-specific instruments used in the Proxy-SVAR draw on variation from geopolitical, OPEC, supply, and transition narratives, all of which achieve reasonable alignment. The speculation category's lower alignment does not compromise identification because the speculation instrument is the strongest individual instrument in both systems ($F = 70.1$ for sentiment, $F = 66.4$ for uncertainty), and the leave-one-out analysis in Appendix C.2 confirms that dropping any single instrument leaves the results unchanged.

Table 8: LLM Validation of Dictionary-Based Classification

Panel A: Oil-Relevance Filter

Metric	Value
Sample size	500
Accuracy	86.4%
Precision	80.0%
Recall	91.7%
F_1 score	85.5%
False positive rate	20.0%

Panel B: Narrative Type Classification (Binary LLM Prompts)

Type	N (dictionary)	LLM agrees	Alignment	FP rate
Geopolitical	80	60	75.0%	25.0%
OPEC	80	64	80.0%	20.0%
Speculation	80	37	46.2%	53.8%
Supply	80	56	70.0%	30.0%
Transition	80	77	96.2%	3.8%

Notes: Panel A reports agreement between the dictionary-based oil-relevance filter and Claude Sonnet 4 on a stratified sample of 500 articles (250 oil-relevant, 250 not). Precision: share of dictionary-flagged articles the LLM also classifies as oil-relevant. Recall: share of LLM-flagged articles the dictionary catches. Panel B reports type-specific validation using binary LLM prompts following Gavriilidis et al. (2026). For each of 500 stratified oil-relevant articles (approximately 80 per specific type), the LLM is asked five separate binary questions corresponding to the five narrative types. Alignment is the share of dictionary-classified articles where the LLM agrees. False positive (FP) rate is the share where the LLM disagrees. Text truncated to first 1,500 words.

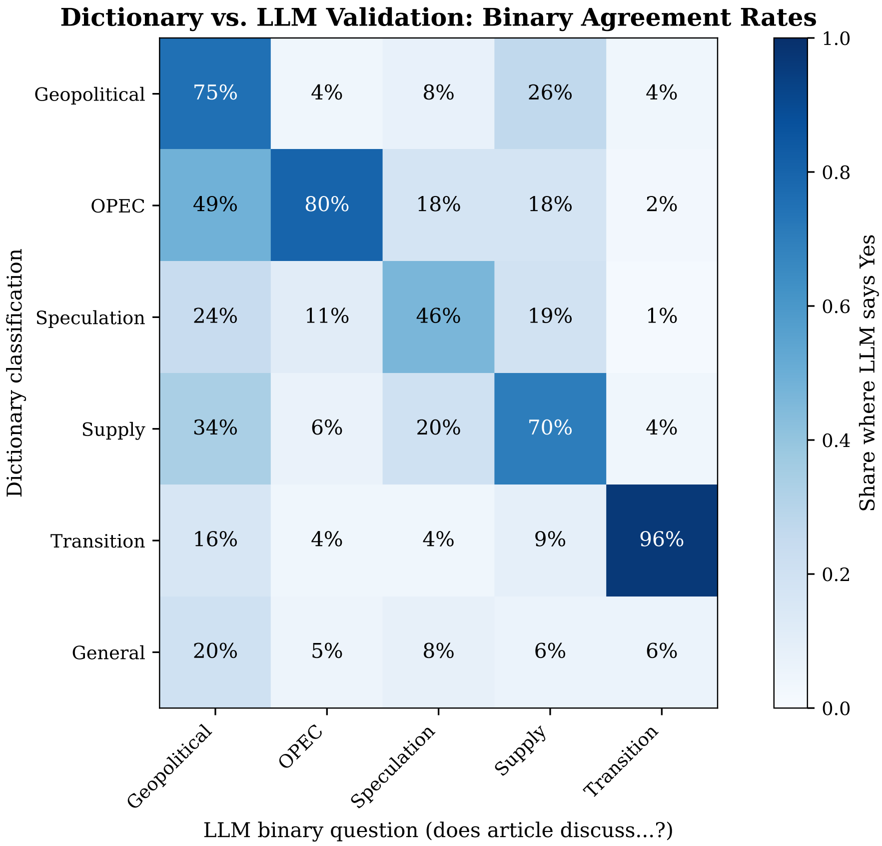


Figure 16: Cross-contamination matrix: dictionary vs. LLM classification. Each cell shows the share of articles classified as the row type by the dictionary that the LLM also flags as the column type (binary yes/no). Diagonal entries correspond to alignment rates reported in Table 8. Off-diagonal entries reflect multi-dimensional narrative content.

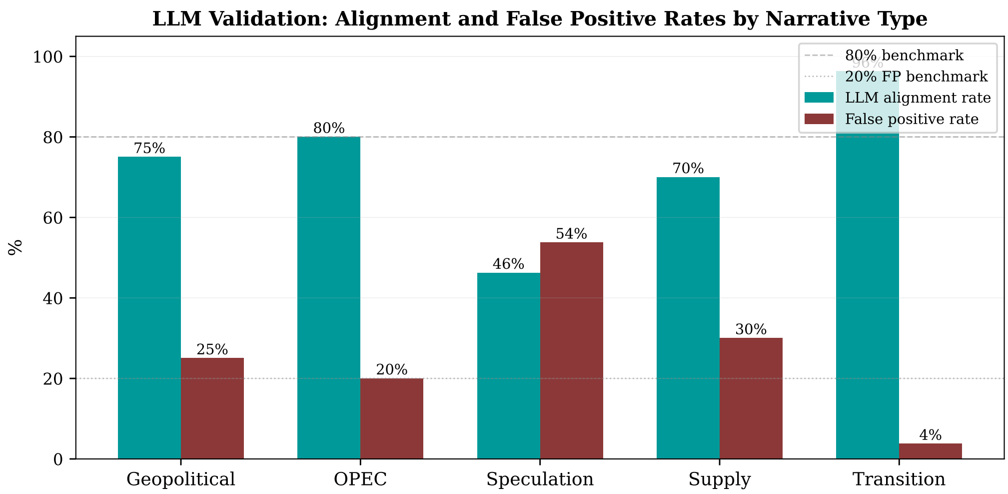
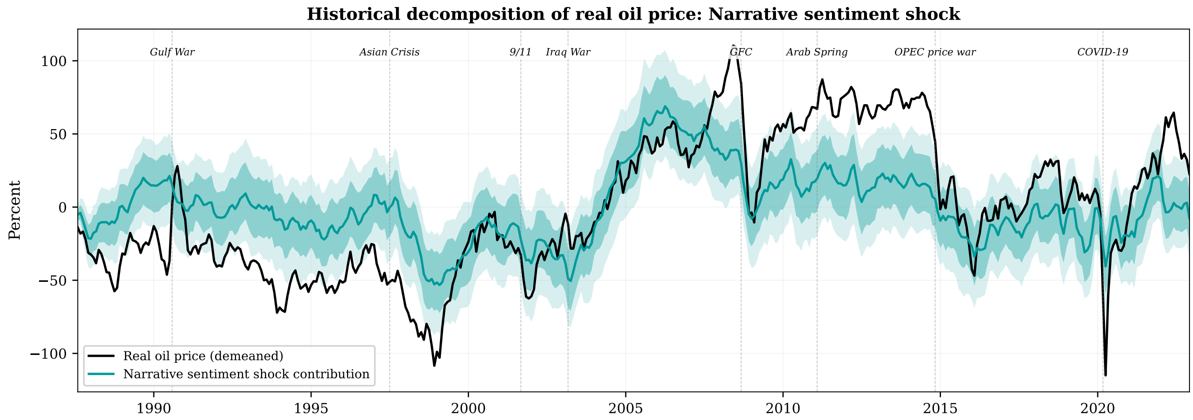


Figure 17: LLM validation: alignment and false positive rates by narrative type. Teal bars: share of dictionary-classified articles where the LLM agrees (alignment rate). Dark red bars: share where the LLM disagrees (false positive rate). Dashed grey lines: 80% alignment and 20% FP benchmarks.

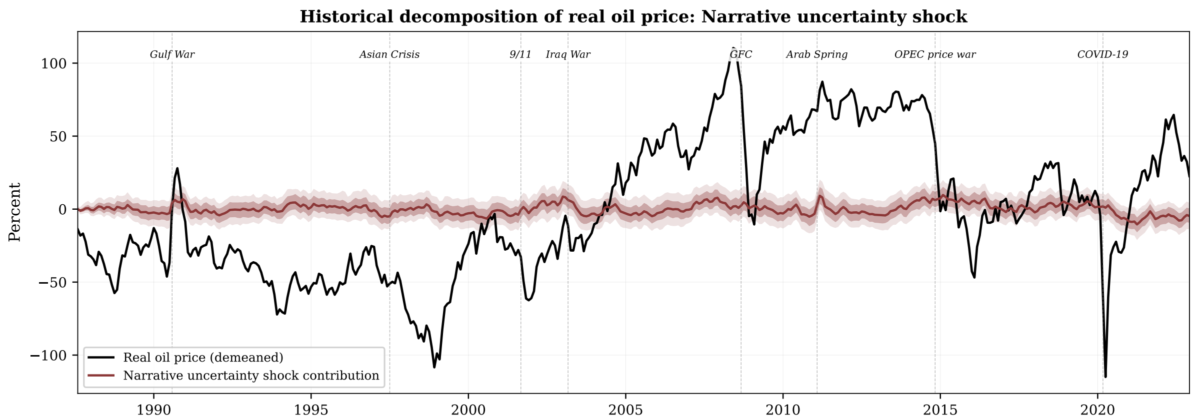
B Additional Results

B.1 Historical Decomposition of Real Oil Price

Figure 18 displays the historical contribution of each narrative shock to the real oil price. The contrast between the two panels encapsulates the paper's central finding. The sentiment shock contribution (Panel a) tracks the demeaned real oil price closely, capturing the sustained bull market of the mid-2000s, the GFC collapse, and the COVID crash. The uncertainty shock contribution (Panel b) is an order of magnitude smaller and shows no systematic tracking of major price movements, confirming that uncertainty operates through the supply channel rather than the price channel.



(a) Narrative sentiment shock



(b) Narrative uncertainty shock

Figure 18: Historical decomposition of real oil price. Solid black line: demeaned real oil price ($100 \times \log(\text{WTI}/\text{CPI})$). Colored line: estimated contribution of each narrative shock. Shaded regions: 68% and 90% MBB confidence bands. Vertical dashed lines mark major oil market events. Sample: 1987:02–2022:12, VAR(6), 10,000 bootstrap replications.

B.2 The Real Options Channel: Rig Count Response

The rig count responses reported in Section 5.3 of the main text are reproduced here for reference alongside a standalone figure isolating the uncertainty shock.

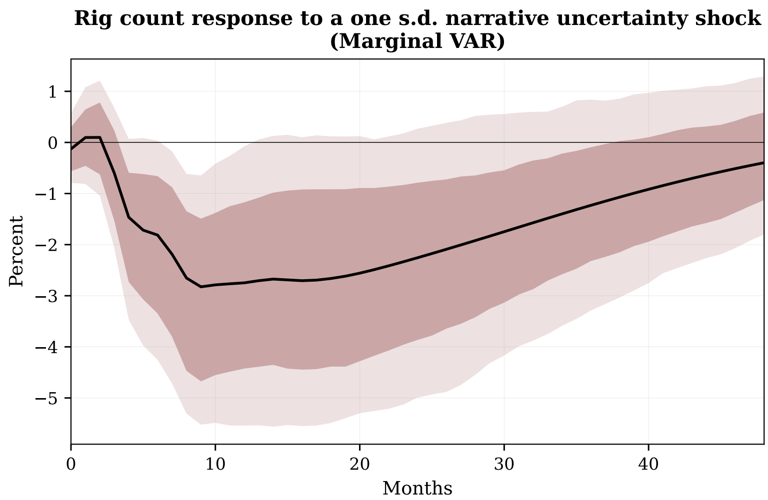


Figure 19: Rig count response to a one standard deviation narrative uncertainty shock (marginal VAR). The rig count declines by 2.8% at the trough (month 9) and remains depressed for over two years, consistent with the real options prediction that heightened ambiguity delays irreversible drilling investment. Dark shaded region: 68% MBB confidence bands. Light shaded region: 90% MBB confidence bands. Sample: 1987:08–2022:12, VAR(6).

C Additional Robustness Checks

C.1 Alternative Lag Order

The baseline VAR uses $p = 6$. Figures 20 and 21 compare the impulse responses from VAR(3), VAR(6), and VAR(12) for the sentiment and uncertainty systems, respectively. The point estimates are qualitatively identical across all three lag orders, with minor differences in confidence band width at longer horizons. The lag order does not affect the paper's conclusions.

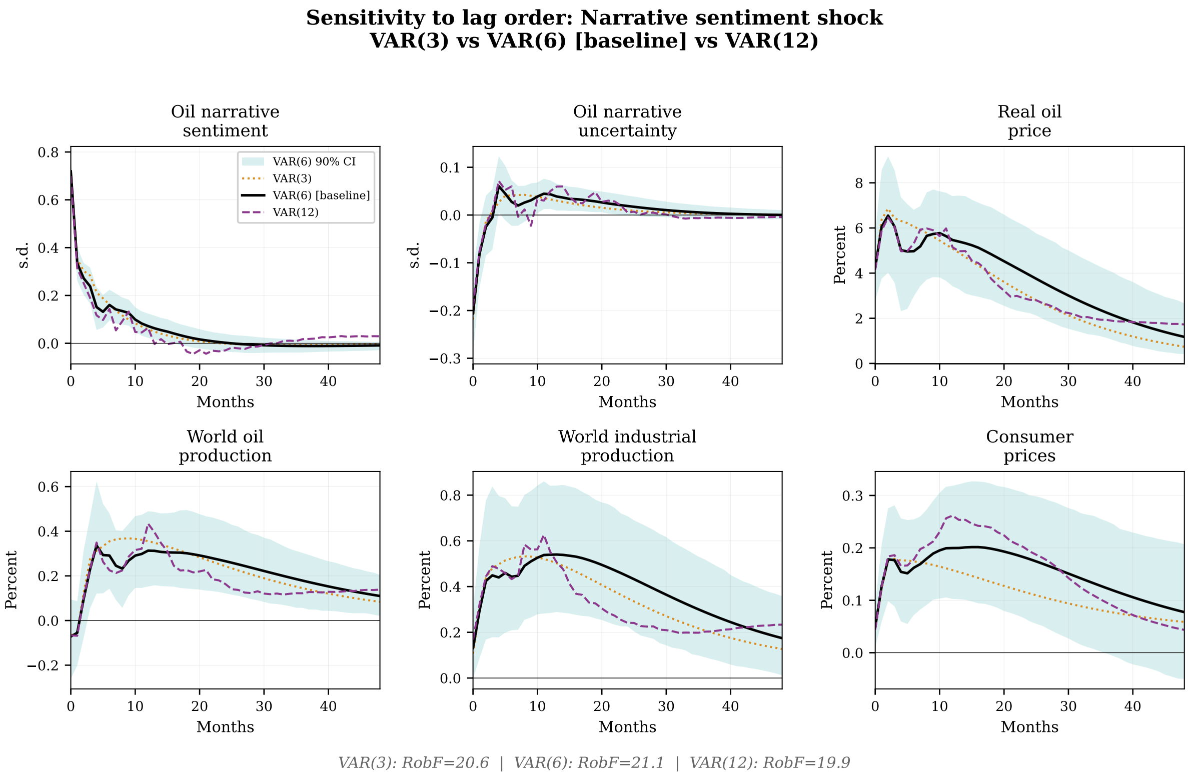


Figure 20: Sensitivity to lag order: narrative sentiment shock. Point estimates from VAR(3) (dotted), VAR(6) (solid, baseline), and VAR(12) (dashed). Shaded region: 90% MBB confidence bands from the baseline VAR(6). Sample: 1987:02–2022:12.

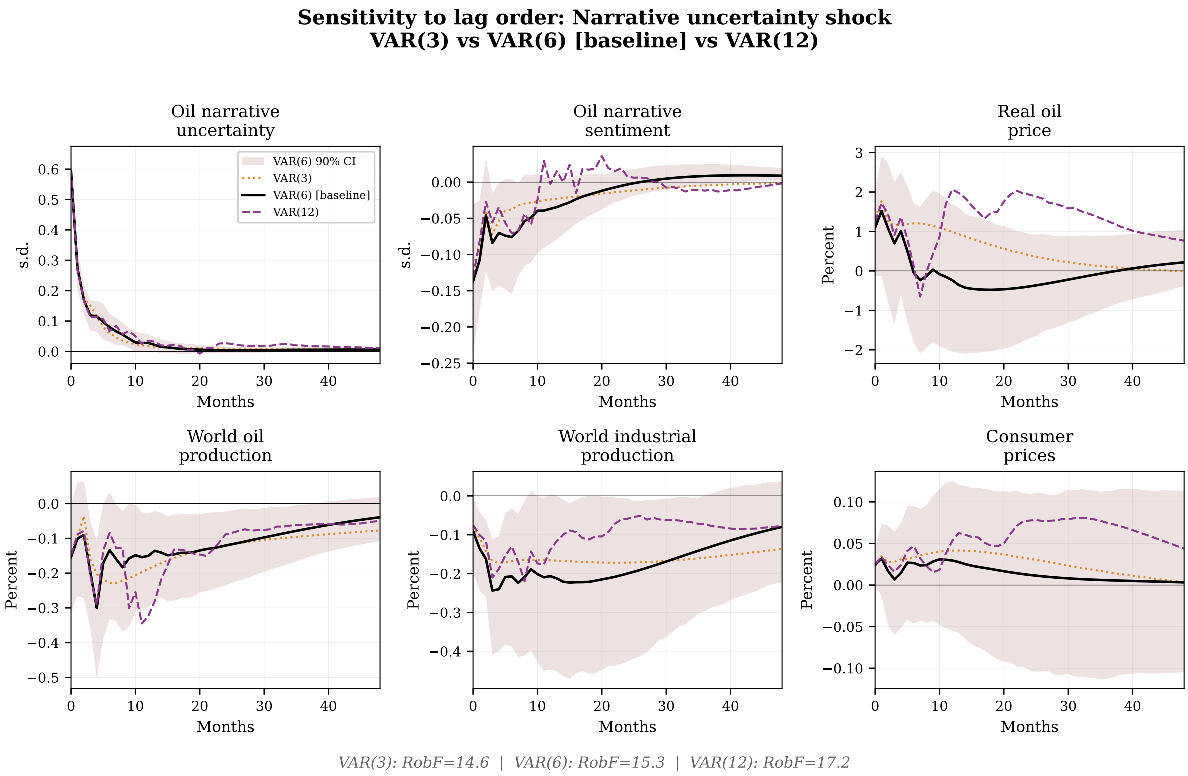


Figure 21: Sensitivity to lag order: narrative uncertainty shock. Point estimates from VAR(3) (dotted), VAR(6) (solid, baseline), and VAR(12) (dashed). Shaded region: 90% MBB confidence bands from the baseline VAR(6). Sample: 1987:02–2022:12.

C.2 Leave-One-Out Instrument Analysis

Figure 22 reports impulse responses from re-estimating the sentiment Proxy-SVAR five times, each time dropping one of the five instruments. In every case, the point estimates remain close to the baseline and within the 90% confidence bands. Dropping speculation (the strongest individual instrument, $F = 70.1$) produces the largest deviation, while dropping supply (the weakest, $F = 4.4$) is virtually invisible. All five leave-one-out specifications maintain joint F-statistics above the Stock-Yogo threshold of 10, ranging from 12.4 (dropping speculation) to 28.1 (dropping supply). This stability confirms that no single instrument drives the results and reinforces the value of the over-identified design.

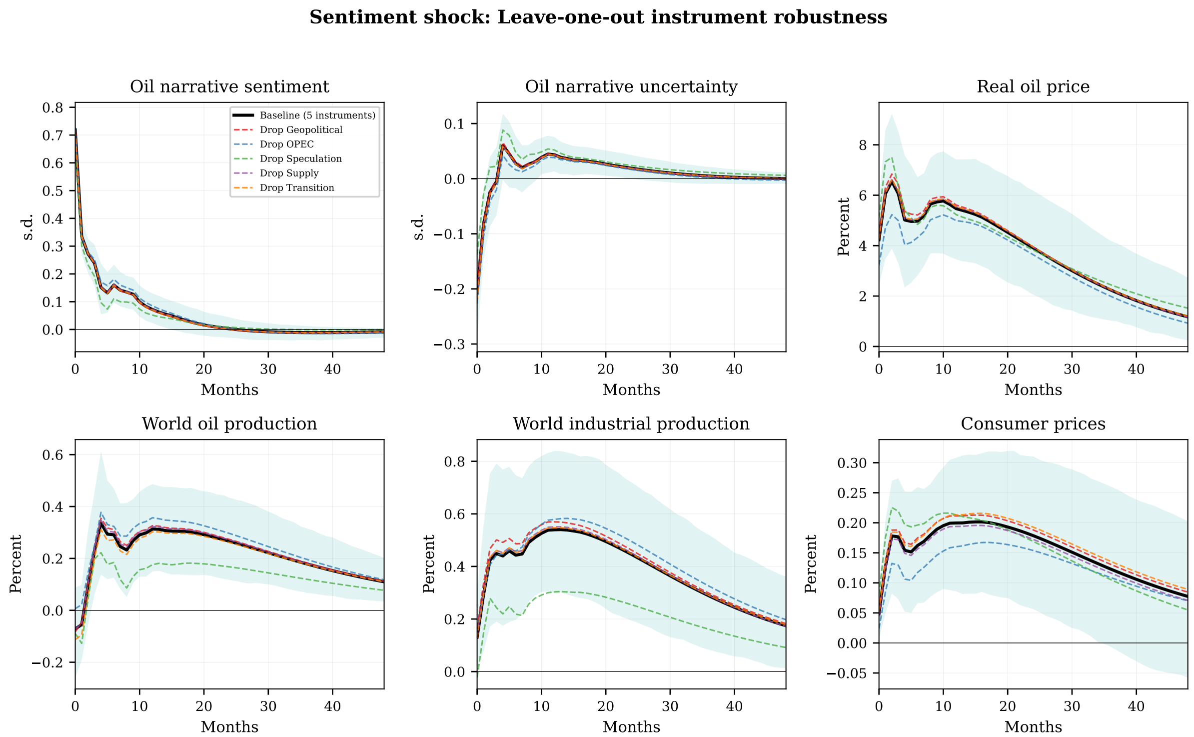


Figure 22: Leave-one-out instrument robustness: narrative sentiment shock. Solid black: baseline with all five instruments. Dashed colored lines: re-estimation dropping one instrument at a time. Shaded region: 90% MBB confidence bands from the baseline. Sample: 1987:02–2022:12, VAR(6).

C.3 Sub-Sample Stability

To assess whether the two-channel result holds across different sample periods, I split the sample at December 1999, separating the pre-financialization era (1987:02–1999:12) from the modern era (2000:01–2022:12) that encompasses the financialization of commodity markets, the rise of index investment, and the emergence of energy transition narratives.

Table 9 reports the first-stage diagnostics for each subsample. The sentiment shock is well-identified in both subsamples: the pre-2000 Robust F is 12.1 and the post-2000 Robust F is 15.5, both above the Stock-Yogo threshold of 10. The uncertainty shock is well-identified only in the post-2000 sample (Robust $F = 12.6$); the pre-2000 first stage is weak (Robust $F = 6.1$), reflecting limited variation in geopolitical and speculation uncertainty instruments during a period with few major ambiguity-generating events (only the Gulf War represents a large uncertainty episode before 2000).

Table 9: First-Stage Diagnostics by Subsample

System	Sample	Robust F	R^2	Status
Sentiment	Full (1987–2022)	21.1	21.5%	Strong
	Pre-2000 (1987–1999)	12.1	28.8%	Strong
	Post-2000 (2000–2022)	15.5	20.4%	Strong
Uncertainty	Full (1987–2022)	15.3	27.4%	Strong
	Pre-2000 (1987–1999)	6.1	26.8%	Weak
	Post-2000 (2000–2022)	12.6	25.5%	Strong

Notes: Robust F-statistics from the first-stage regression of the reduced-form VAR residual on five purged type-specific instruments. "Strong" denotes Robust $F > 10$ (Stock-Yogo threshold). VAR(6) in all specifications.

Figures 23 and 24 compare the impulse responses across subsamples.

Sentiment shock. The demand-side information channel operates in both regimes. The real oil price response is positive in both subsamples, with a substantially larger magnitude in the post-2000 sample (+8.6% peak vs. +2.7% pre-2000), consistent with financialization amplifying the information channel through which narrative sentiment transmits to oil prices. World industrial production and consumer prices respond positively in the post-2000 sample; the pre-2000 responses are smaller and less precise, reflecting the shorter sample and lower financialization. The qualitative conclusion—bullish narrative sentiment raises oil prices and activity—holds across both eras.

Uncertainty shock. The post-2000 subsample, where the first stage is strong (Robust $F = 12.6$), confirms the full-sample findings: the uncertainty shock reduces world oil production (-0.48%) and world industrial production (-0.31%), with a positive but imprecise oil price response. These magnitudes are larger than the full-sample estimates, consistent with the real options channel operating more strongly in the post-financialization regime, where geopolitical and speculative uncertainty are more prominent. The pre-2000 uncertainty results should be interpreted with caution given the weak first stage ($F = 6.1$).

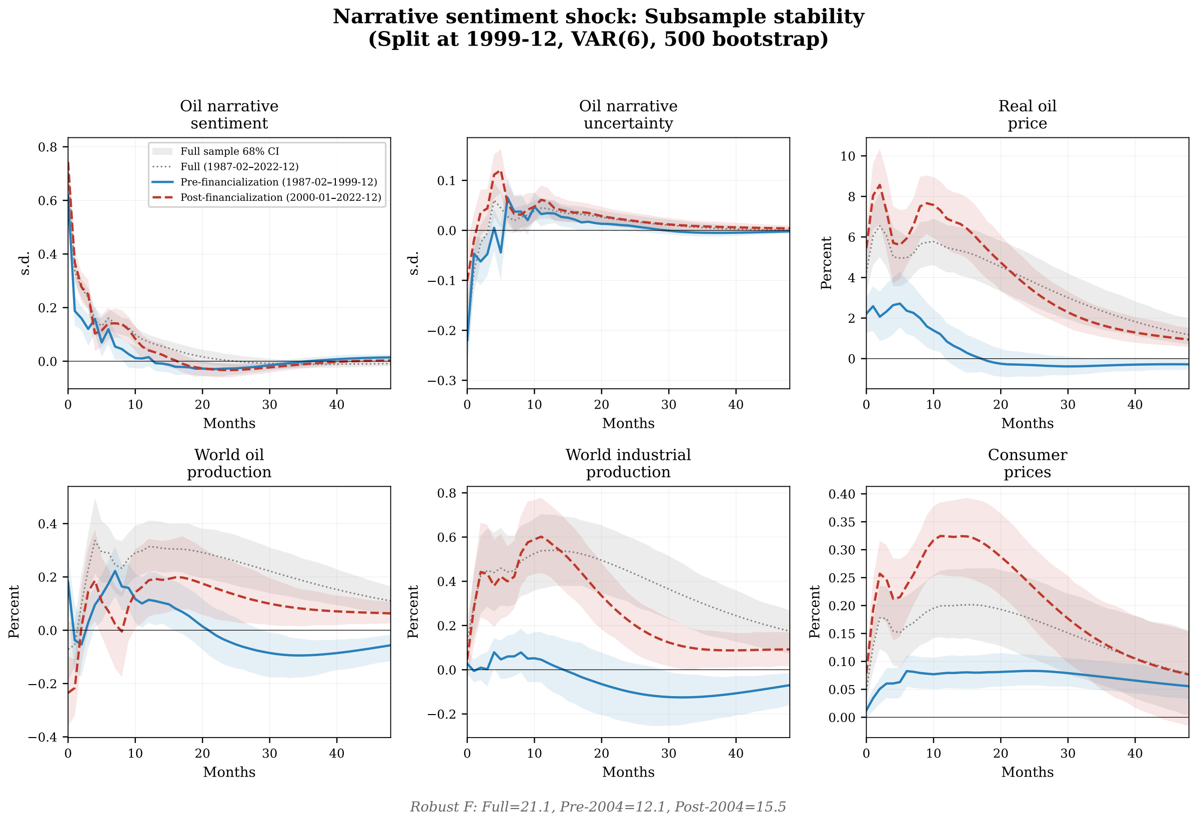


Figure 23: Sub-sample stability: narrative sentiment shock. Grey dotted line with shaded 68% band: full-sample baseline (1987–2022). Blue solid: pre-2000 (1987–1999) with 68% band. Red dashed: post-2000 (2000–2022) with 68% band. The oil price response is present in both subsamples but larger post-2000, consistent with financialization amplifying the demand-side information channel. VAR(6).

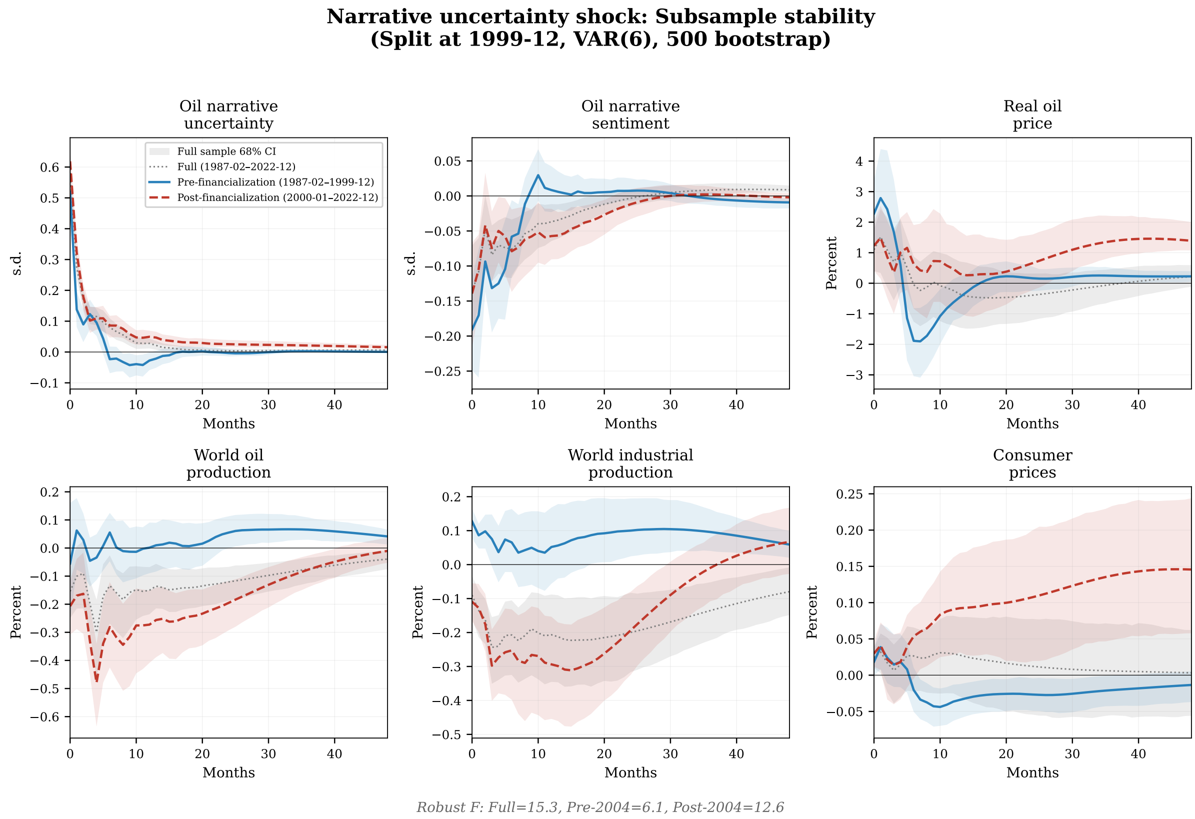
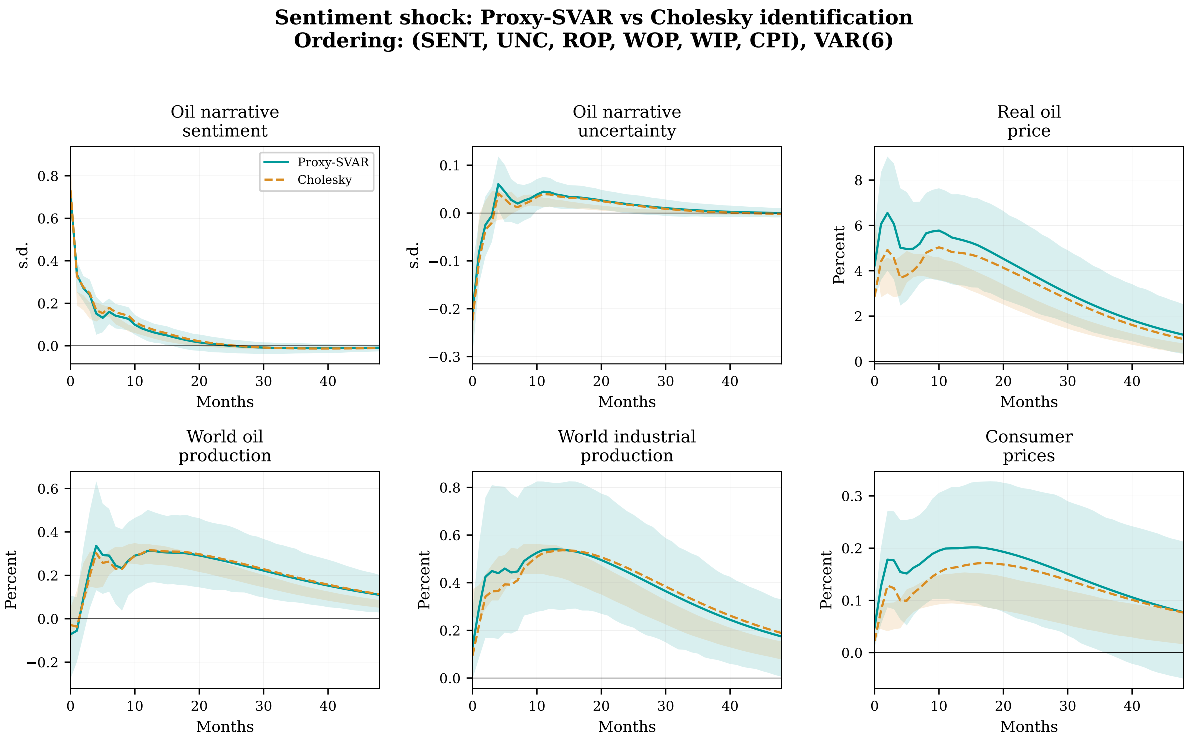


Figure 24: Sub-sample stability: narrative uncertainty shock. Grey dotted line with shaded 68% band: full-sample baseline (1987–2022). Blue solid: pre-2000 (1987–1999) with 68% band. Red dashed: post-2000 (2000–2022) with 68% band. The production decline is confirmed in the post-2000 sample where the first stage is strong (Robust $F = 12.6$). Pre-2000 results should be interpreted with caution (Robust $F = 6.1$). VAR(6).

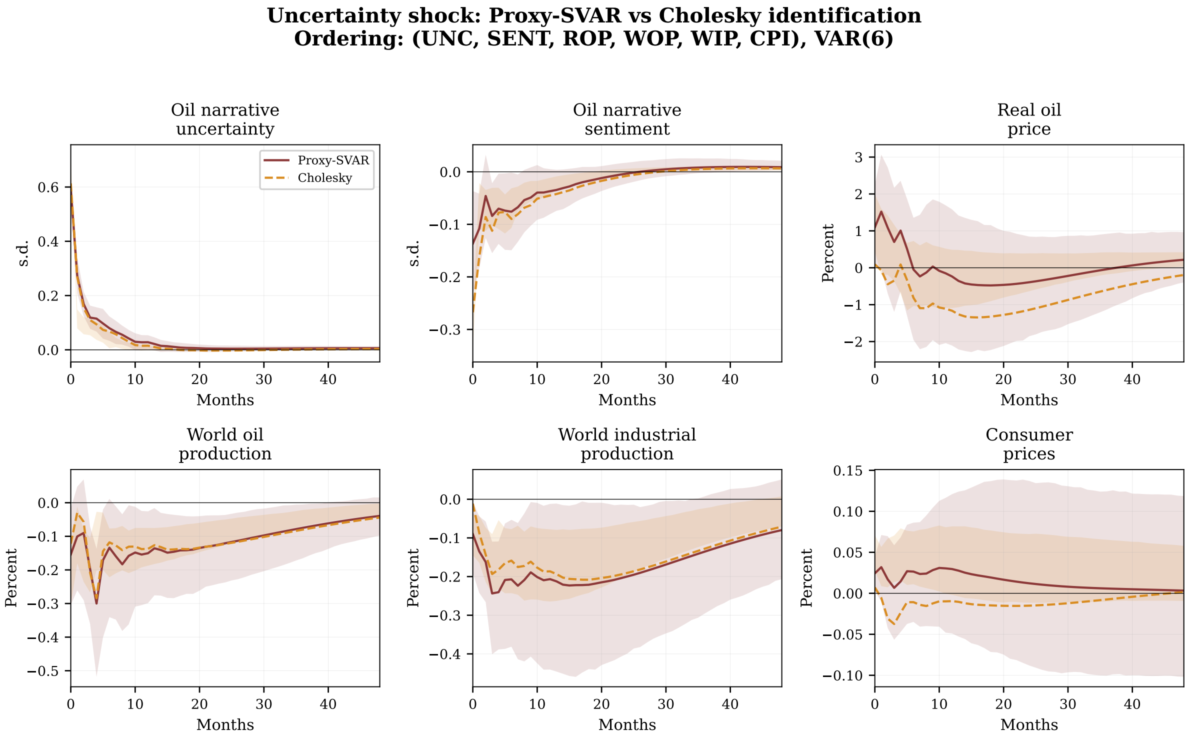
C.4 Cholesky Identification

As a benchmark, I estimate the VAR with recursive (Cholesky) identification using the ordering (SENT, UNC, ROP, WOP, WIP, CPI), where the narrative variables are ordered first under the assumption that they respond to macroeconomic developments with a one-month lag. This approach, standard in the policy uncertainty literature (Baker et al., 2016), identifies shocks directly from the reduced-form residuals under short-run timing restrictions rather than using external instruments.

The Cholesky impulse responses are qualitatively similar to the Proxy-SVAR results for the sentiment shock but less precisely estimated for the uncertainty shock, where the recursive ordering assumption is harder to justify—there is no strong reason to assume that uncertainty responds to oil prices only with a lag, and a different ordering could reverse the sign of the price response. The Proxy-SVAR is preferred because it does not require an ordering assumption and leverages the external instrument variation for identification. The close correspondence between the two approaches for the sentiment shock, which rely on different sources of identifying variation, lends further credibility to the identified narrative sentiment shock.



(a) Narrative sentiment shock



(b) Narrative uncertainty shock

Figure 25: Cholesky vs. Proxy-SVAR identification. Solid colored line: Proxy-SVAR baseline with 90% MBB confidence bands (shaded). Dashed orange: Cholesky identification with 68% MBB confidence bands (shaded). Panel (a): sentiment shock with ordering (SENT, UNC, ROP, WOP, WIP, CPI). Panel (b): uncertainty shock with ordering (UNC, SENT, ROP, WOP, WIP, CPI). The Cholesky and Proxy-SVAR point estimates agree closely for the sentiment shock. For the uncertainty shock, the Cholesky estimates are less precisely estimated, reflecting the sensitivity of recursive identification to the ordering assumption. Sample: 1987:02–2022:12, VAR(6).